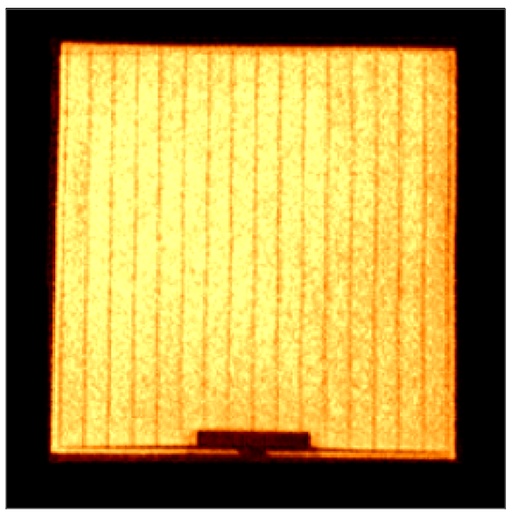
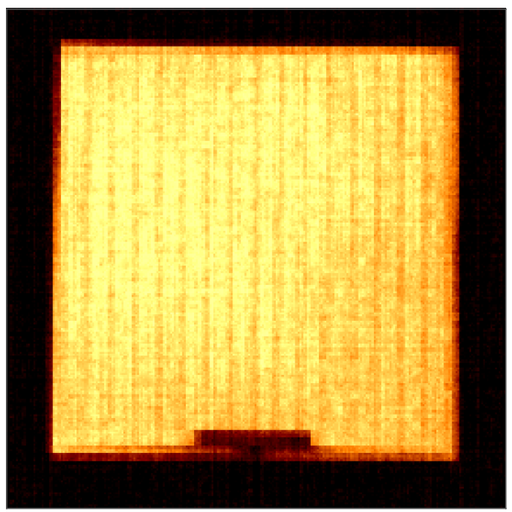
Bad images for UTV | Reference_0021

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

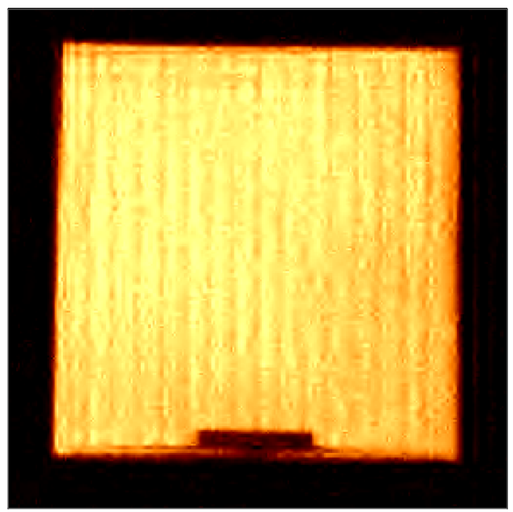
Ground truth
Ground truth



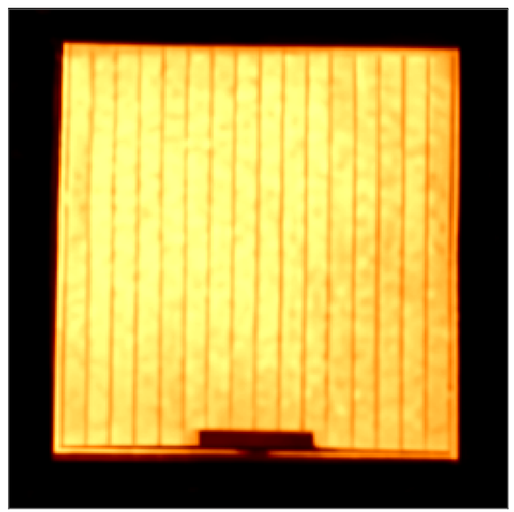
Zero-fill
PSNR 24.47 dB | SSIM 0.509
HaarPSI 0.564



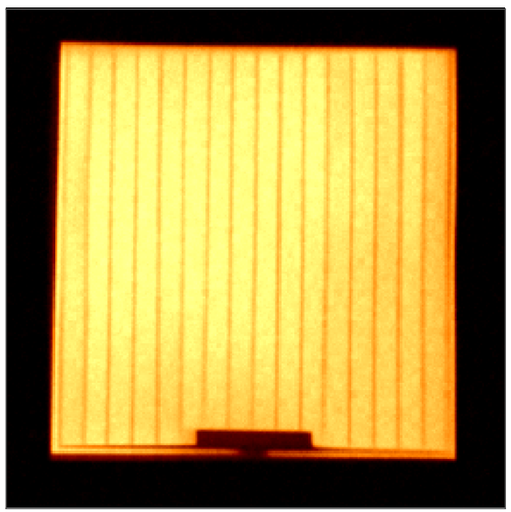
SPGL1-BPDN
PSNR 26.06 dB | SSIM 0.580
HaarPSI 0.591



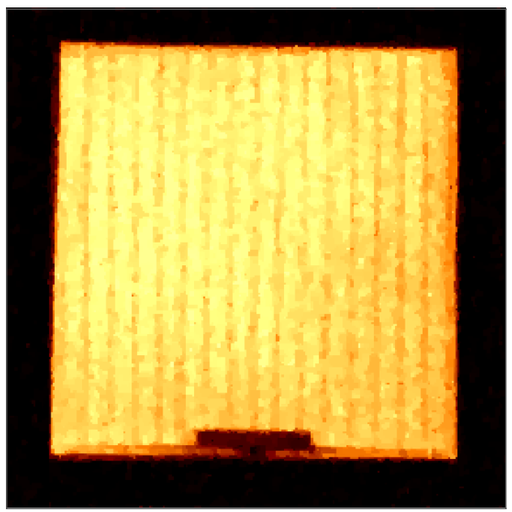
PnP-ADMM-tuned-5
PSNR 31.07 dB | SSIM 0.847
HaarPSI 0.822



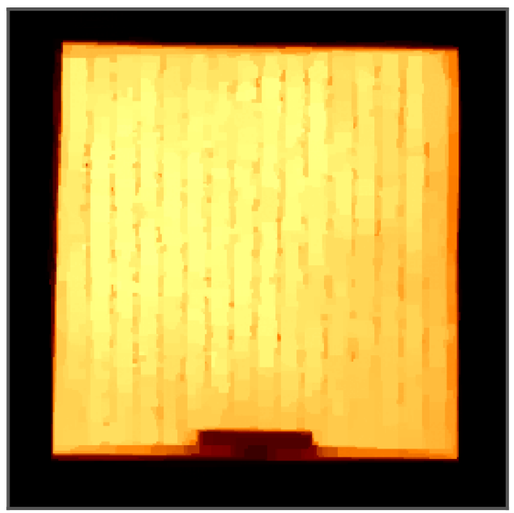
GS-PnP-tuned-5
PSNR 28.76 dB | SSIM 0.705
HaarPSI 0.755



Scalar TV
PSNR 27.48 dB | SSIM 0.642
HaarPSI 0.692



UTV C60
PSNR 27.83 dB | SSIM 0.695
HaarPSI 0.574





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

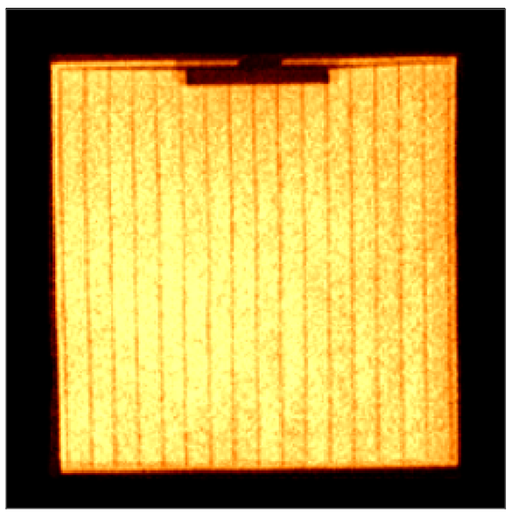
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

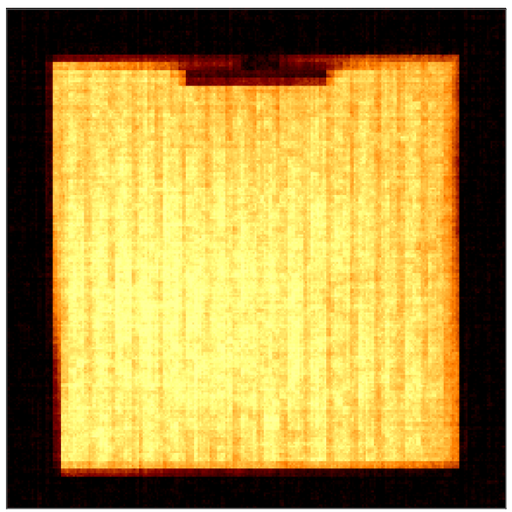
Bad images for UTV | Reference_0066

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

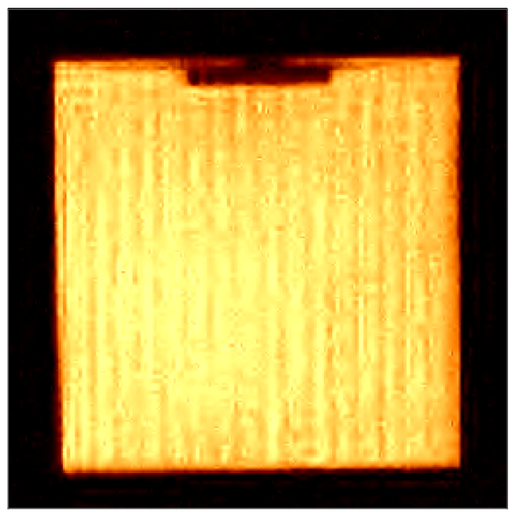
Ground truth
Ground truth



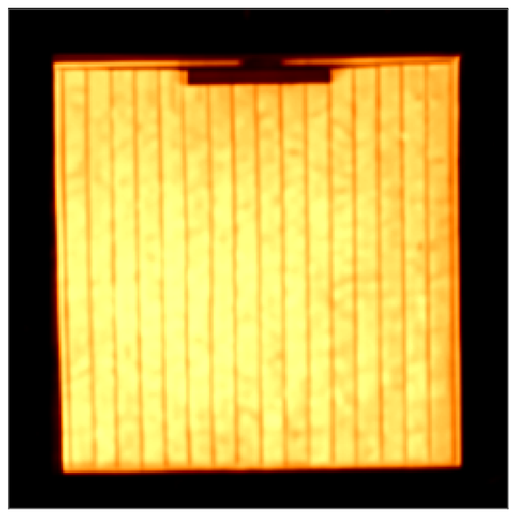
Zero-fill
PSNR 24.59 dB | SSIM 0.516
HaarPSI 0.544



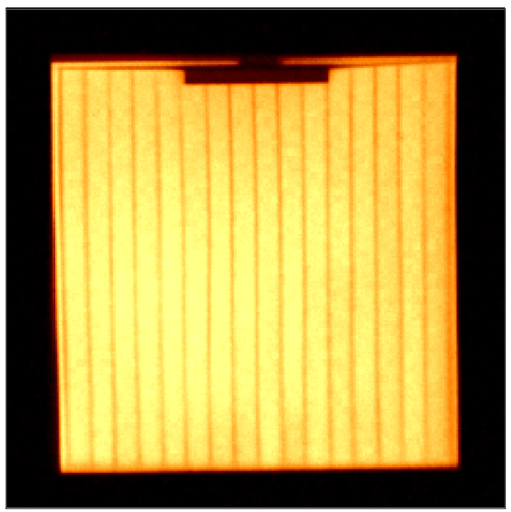
SPGL1-BPDN
PSNR 26.25 dB | SSIM 0.579
HaarPSI 0.599



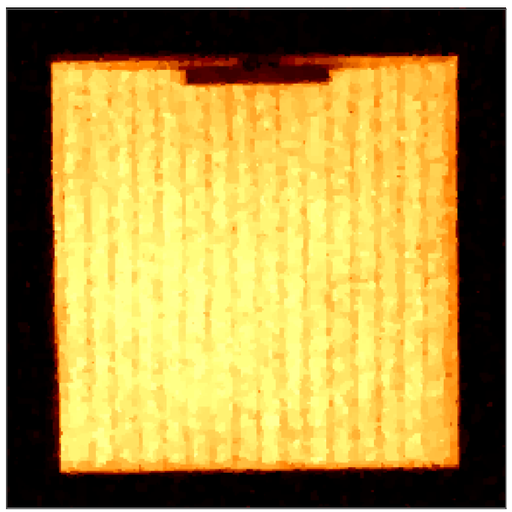
PnP-ADMM-tuned-5
PSNR 31.05 dB | SSIM 0.846
HaarPSI 0.814



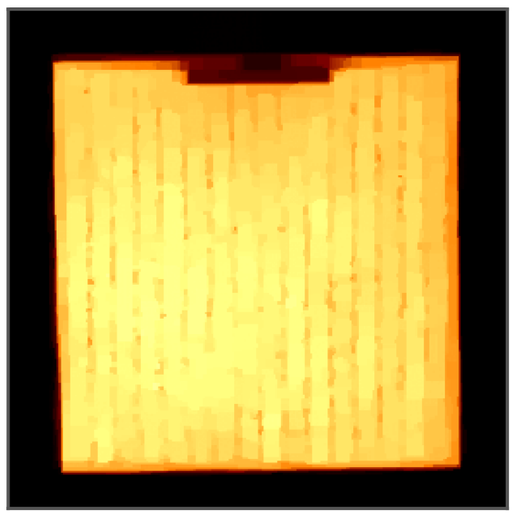
GS-PnP-tuned-5
PSNR 28.70 dB | SSIM 0.702
HaarPSI 0.748



Scalar TV
PSNR 27.54 dB | SSIM 0.650
HaarPSI 0.684



UTV C60
PSNR 27.67 dB | SSIM 0.688
HaarPSI 0.568





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

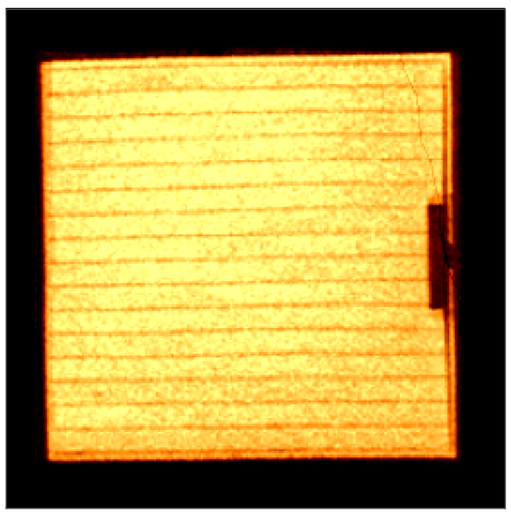
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

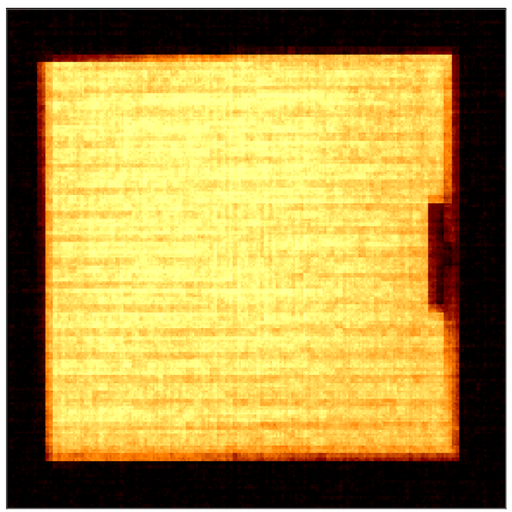
Bad images for UTV | Reference_0188

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

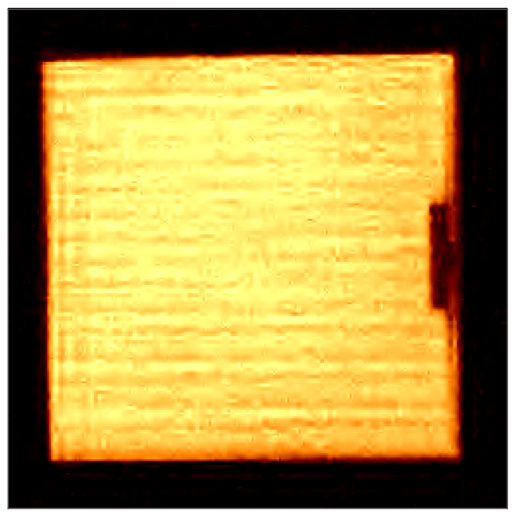
Ground truth
Ground truth



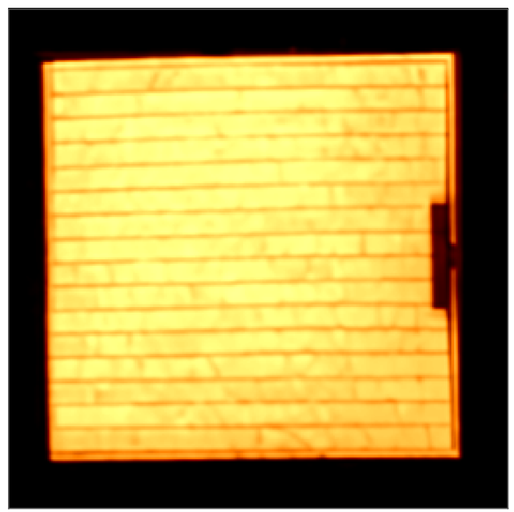
Zero-fill
PSNR 24.51 dB | SSIM 0.510
HaarPSI 0.565



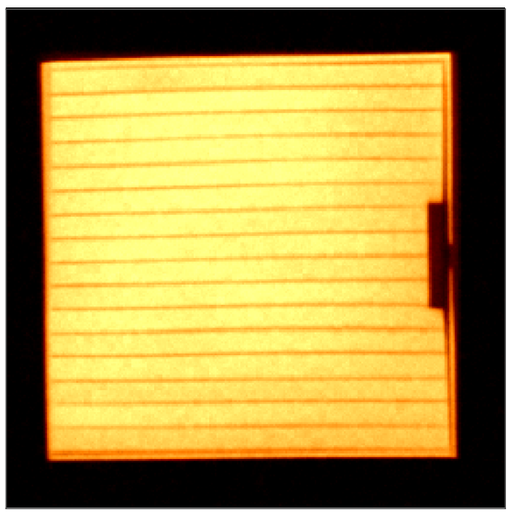
SPGL1-BPDN
PSNR 26.02 dB | SSIM 0.573
HaarPSI 0.606



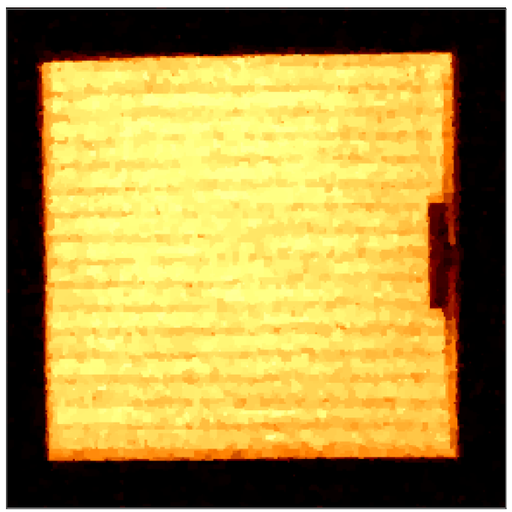
PnP-ADMM-tuned-5
PSNR 30.84 dB | SSIM 0.838
HaarPSI 0.818



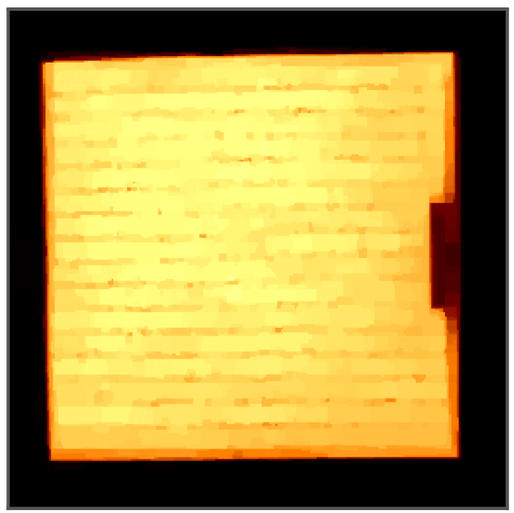
GS-PnP-tuned-5
PSNR 28.54 dB | SSIM 0.698
HaarPSI 0.744



Scalar TV
PSNR 27.35 dB | SSIM 0.643
HaarPSI 0.685



UTV C60
PSNR 27.74 dB | SSIM 0.700
HaarPSI 0.596





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

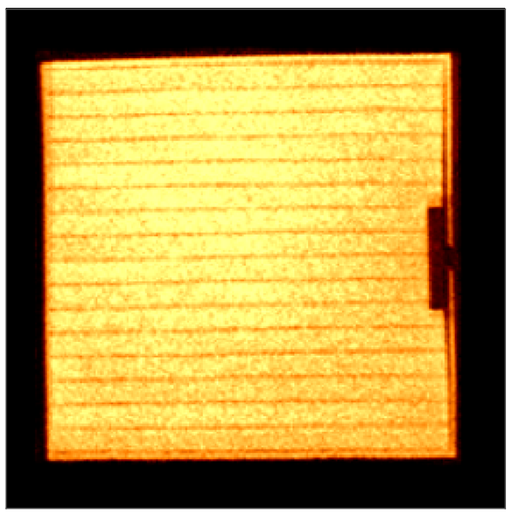
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

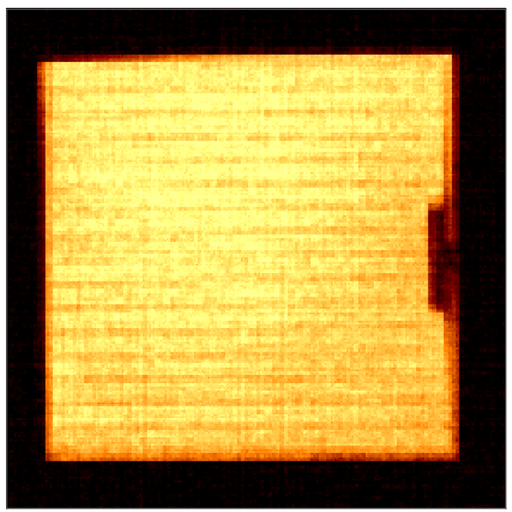
Bad images for UTV | Reference_0199

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

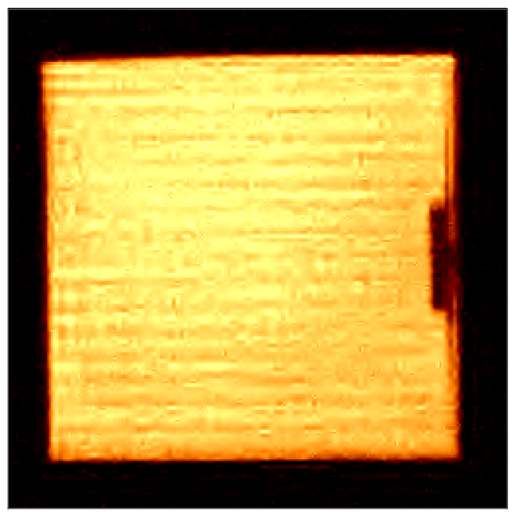
Ground truth
Ground truth



Zero-fill
PSNR 24.70 dB | SSIM 0.506
HaarPSI 0.578



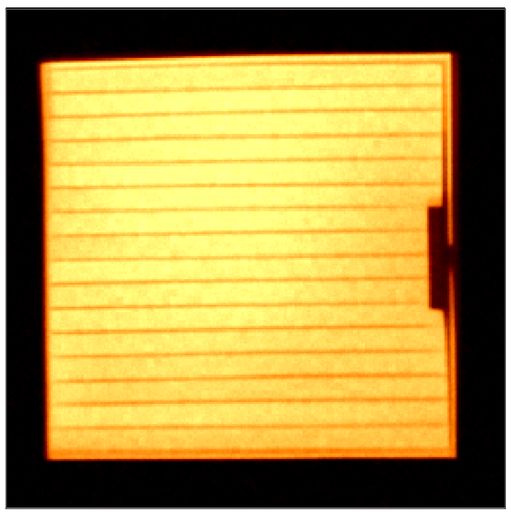
SPGL1-BPDN
PSNR 26.21 dB | SSIM 0.579
HaarPSI 0.598



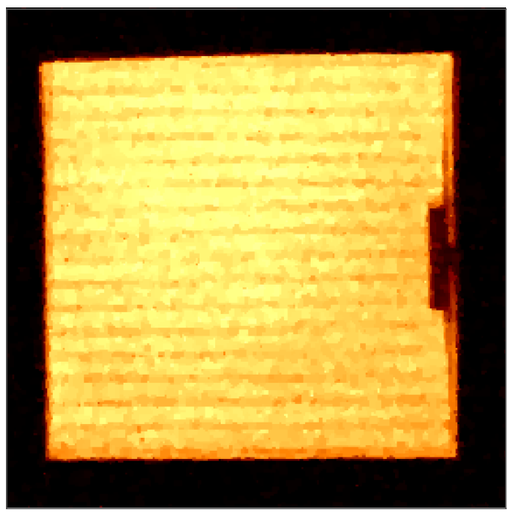
PnP-ADMM-tuned-5
PSNR 30.96 dB | SSIM 0.839
HaarPSI 0.819



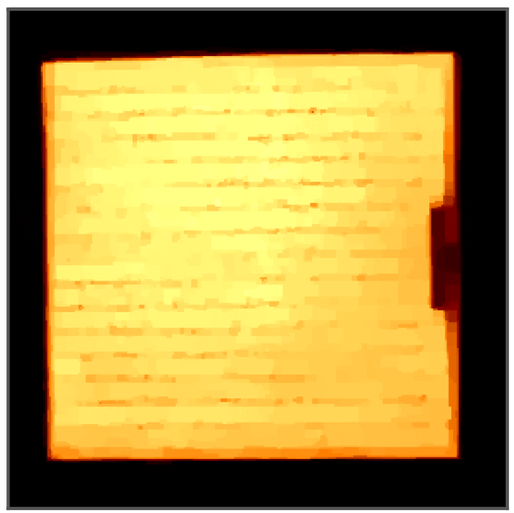
GS-PnP-tuned-5
PSNR 28.58 dB | SSIM 0.695
HaarPSI 0.747



Scalar TV
PSNR 27.56 dB | SSIM 0.640
HaarPSI 0.693



UTV C60
PSNR 27.77 dB | SSIM 0.688
HaarPSI 0.579





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

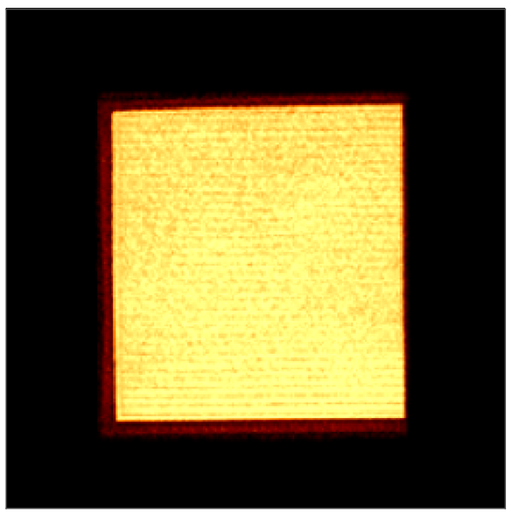
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

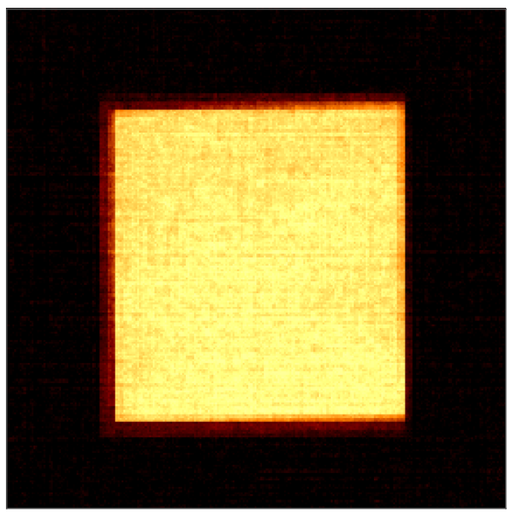
Bad images for UTV | Reference_0260

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

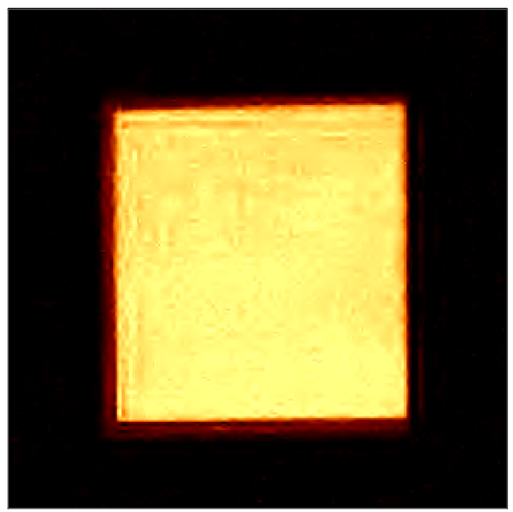
Ground truth
Ground truth



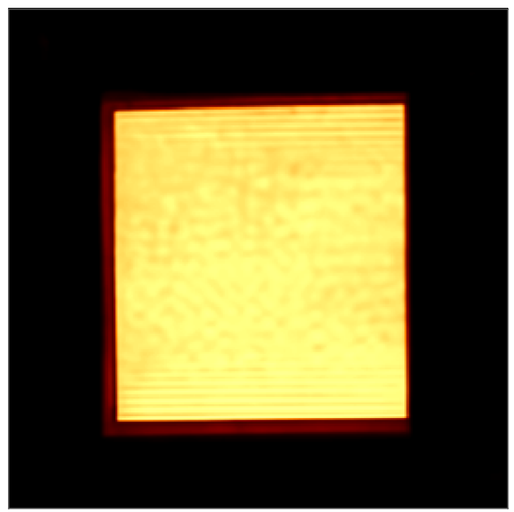
Zero-fill
PSNR 25.97 dB | SSIM 0.442
HaarPSI 0.532



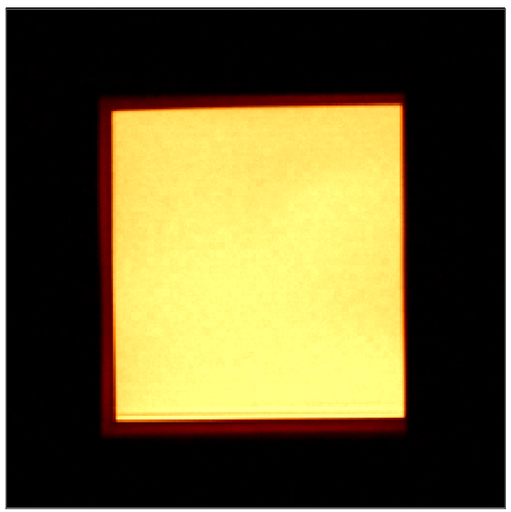
SPGL1-BPDN
PSNR 29.25 dB | SSIM 0.679
HaarPSI 0.618



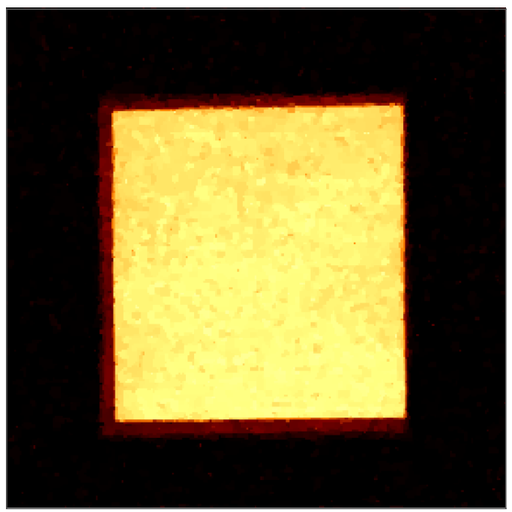
PnP-ADMM-tuned-5
PSNR 33.76 dB | SSIM 0.860
HaarPSI 0.825



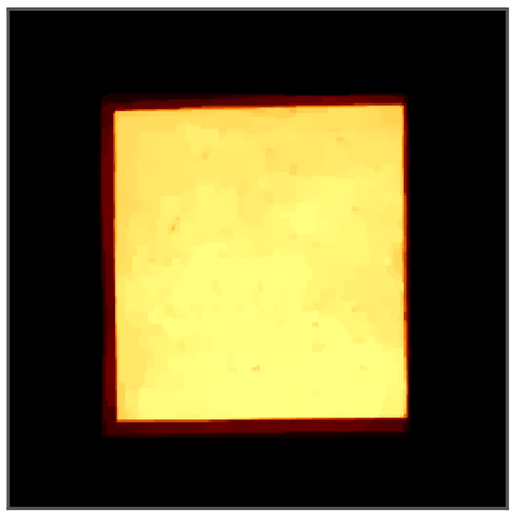
GS-PnP-tuned-5
PSNR 31.70 dB | SSIM 0.736
HaarPSI 0.733



Scalar TV
PSNR 30.61 dB | SSIM 0.638
HaarPSI 0.746



UTV C60
PSNR 32.38 dB | SSIM 0.791
HaarPSI 0.650





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

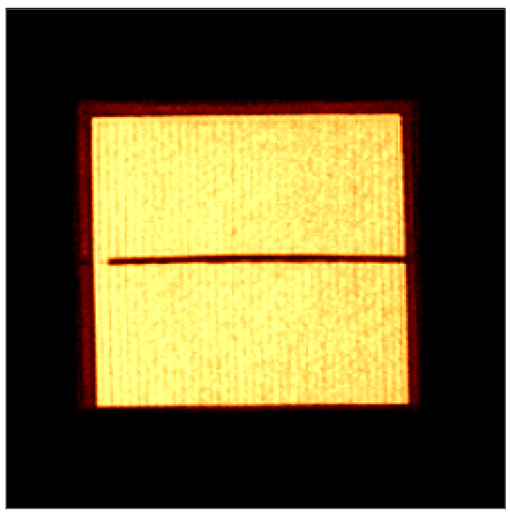
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Visual observation: all compared methods failed to recover the lines.

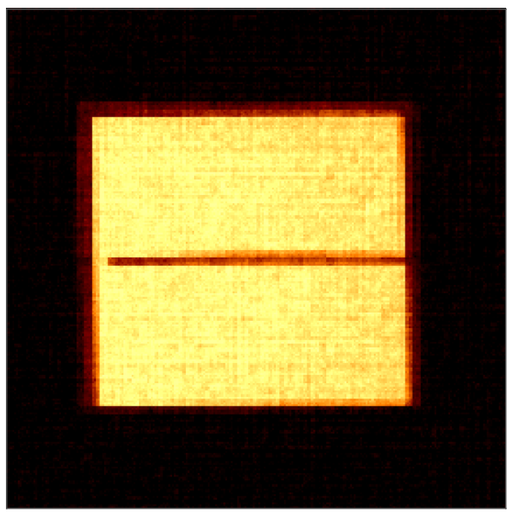
Bad images for UTV | Reference_0357

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

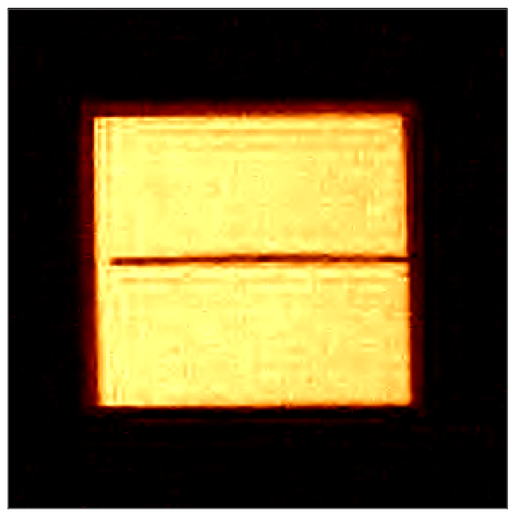
Ground truth
Ground truth



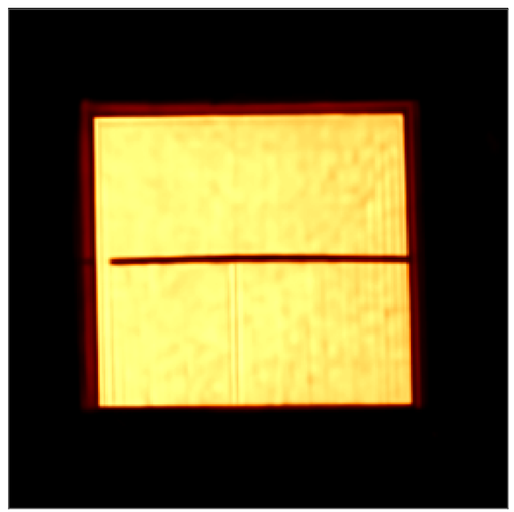
Zero-fill
PSNR 24.75 dB | SSIM 0.432
HaarPSI 0.526



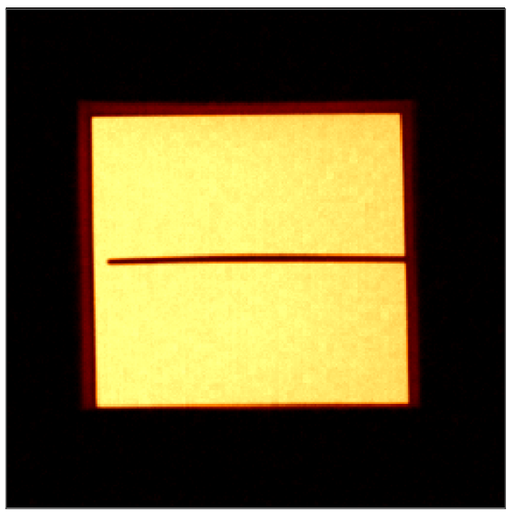
SPGL1-BPDN
PSNR 27.79 dB | SSIM 0.664
HaarPSI 0.585



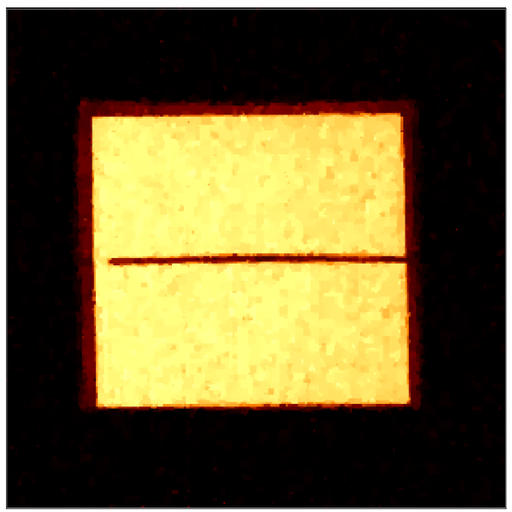
PnP-ADMM-tuned-5
PSNR 33.46 dB | SSIM 0.864
HaarPSI 0.806



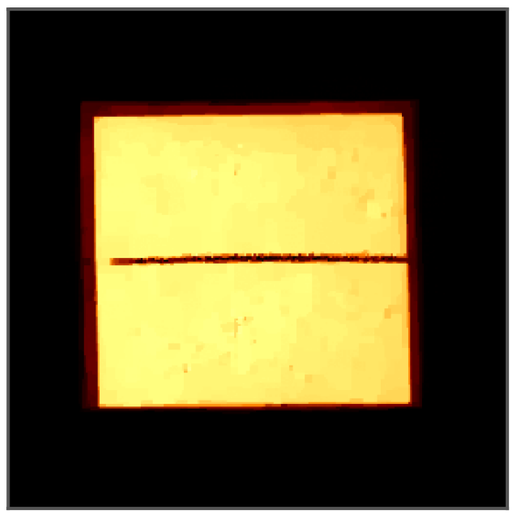
GS-PnP-tuned-5
PSNR 30.78 dB | SSIM 0.683
HaarPSI 0.715



Scalar TV
PSNR 30.04 dB | SSIM 0.649
HaarPSI 0.737



UTV C60
PSNR 31.50 dB | SSIM 0.802
HaarPSI 0.685





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

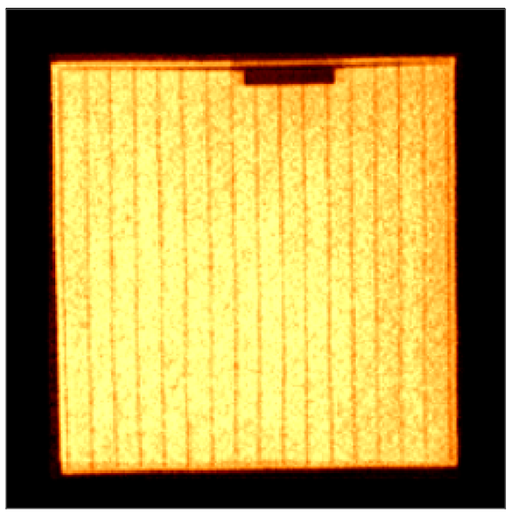
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

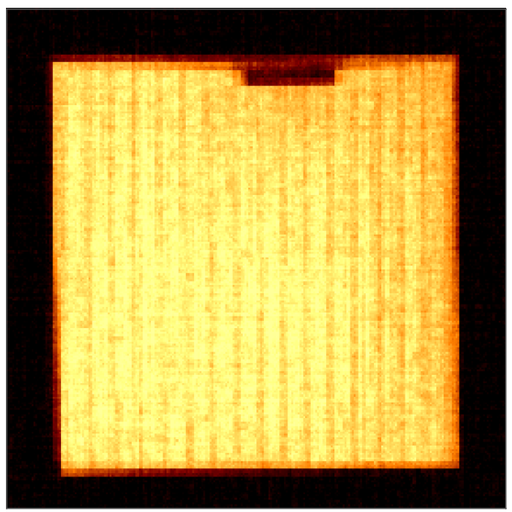
Bad images for UTV | Reference_0727

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

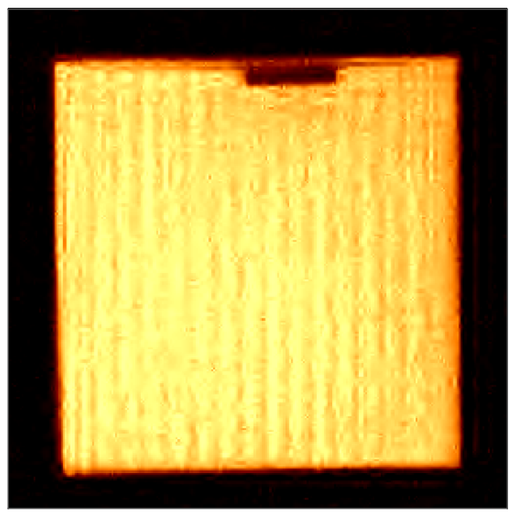
Ground truth
Ground truth



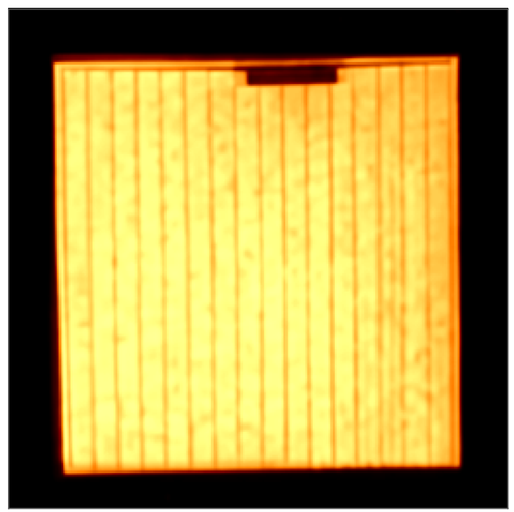
Zero-fill
PSNR 24.59 dB | SSIM 0.511
HaarPSI 0.574



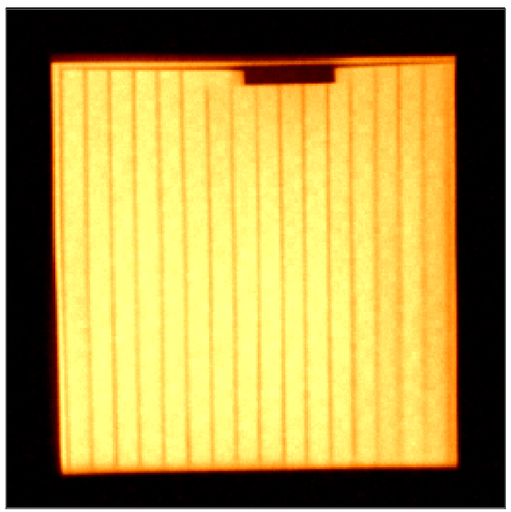
SPGL1-BPDN
PSNR 26.02 dB | SSIM 0.578
HaarPSI 0.607



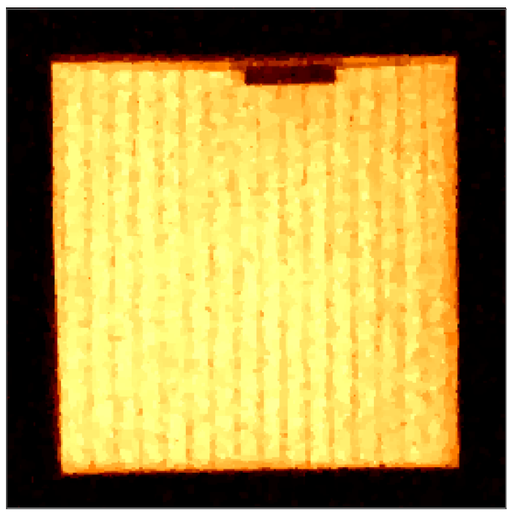
PnP-ADMM-tuned-5
PSNR 31.04 dB | SSIM 0.845
HaarPSI 0.828



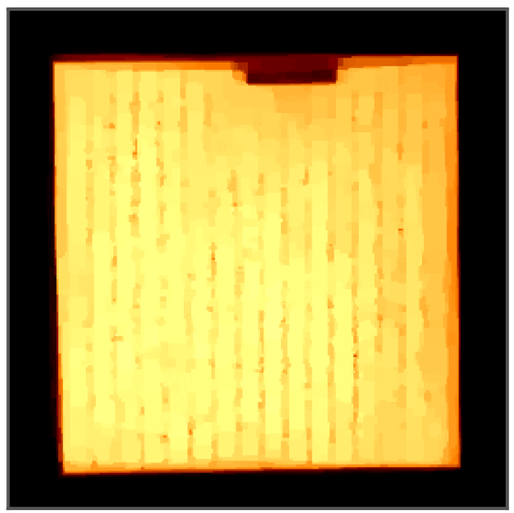
GS-PnP-tuned-5
PSNR 28.67 dB | SSIM 0.695
HaarPSI 0.752



Scalar TV
PSNR 27.44 dB | SSIM 0.647
HaarPSI 0.677



UTV C60
PSNR 27.83 dB | SSIM 0.698
HaarPSI 0.600





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

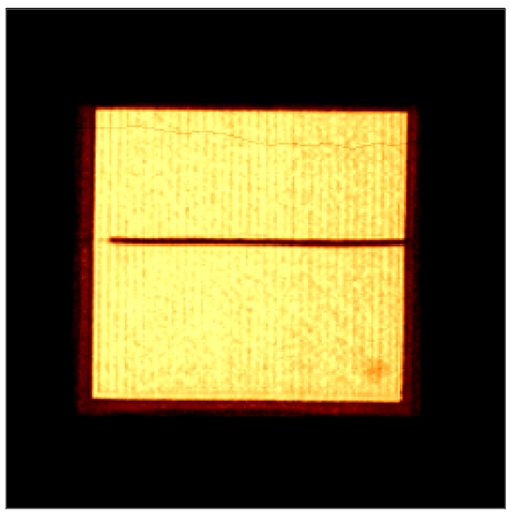
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

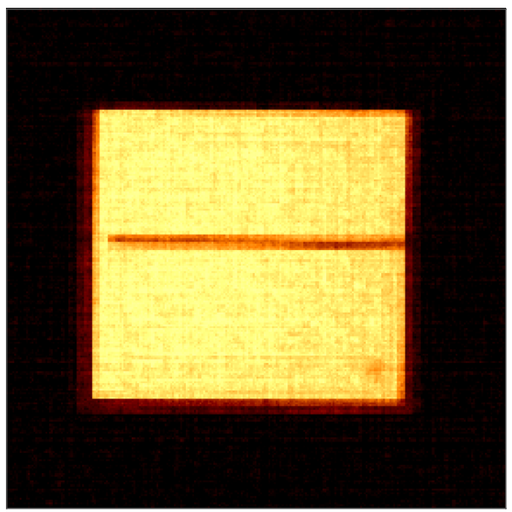
Bad images for UTV | Reference_0793

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

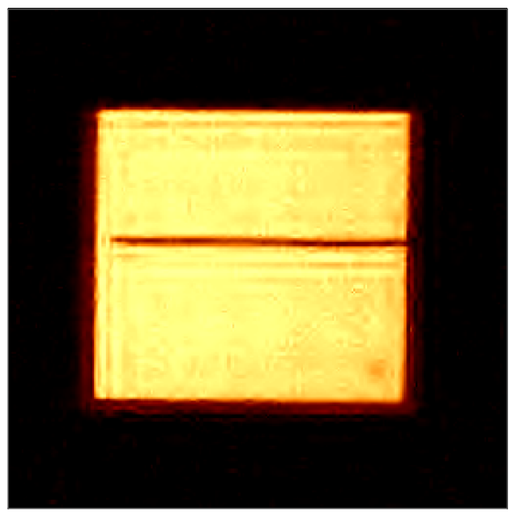
Ground truth
Ground truth



Zero-fill
PSNR 24.50 dB | SSIM 0.416
HaarPSI 0.501



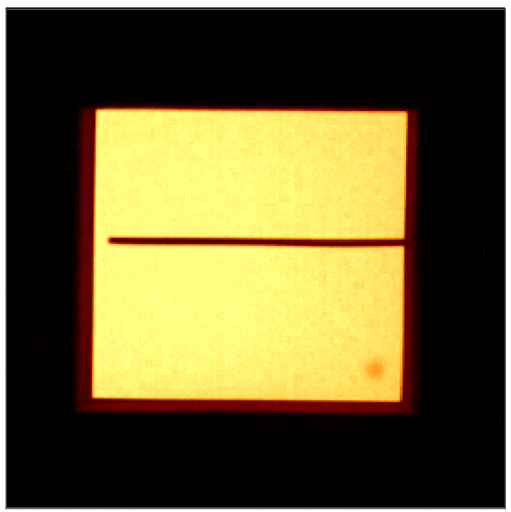
SPGL1-BPDN
PSNR 28.04 dB | SSIM 0.669
HaarPSI 0.580



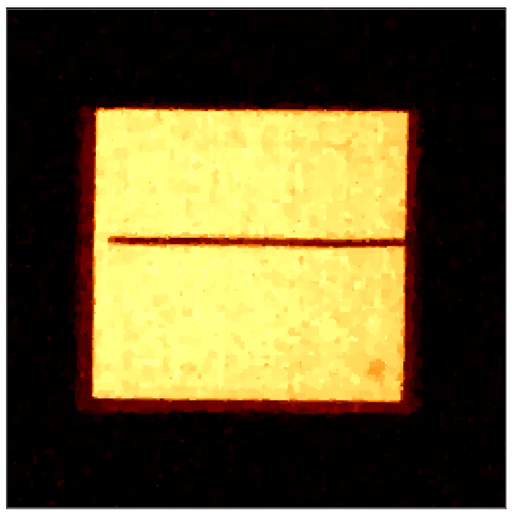
PnP-ADMM-tuned-5
PSNR 33.17 dB | SSIM 0.854
HaarPSI 0.778



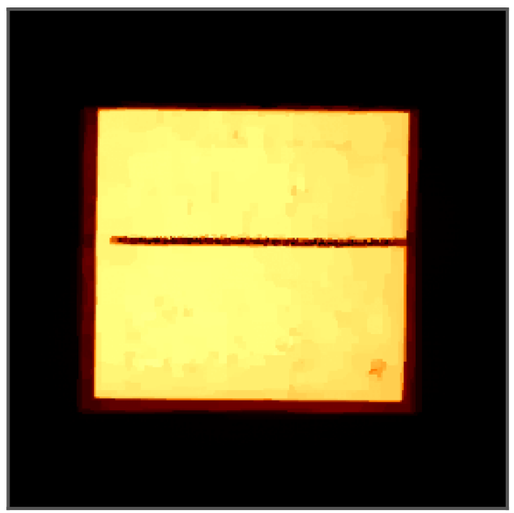
GS-PnP-tuned-5
PSNR 30.74 dB | SSIM 0.679
HaarPSI 0.718



Scalar TV
PSNR 29.78 dB | SSIM 0.634
HaarPSI 0.715



UTV C60
PSNR 31.46 dB | SSIM 0.782
HaarPSI 0.683





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

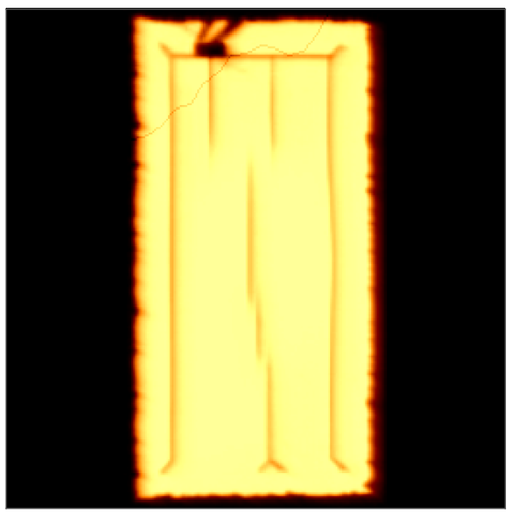
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Visual observation: all compared methods failed.

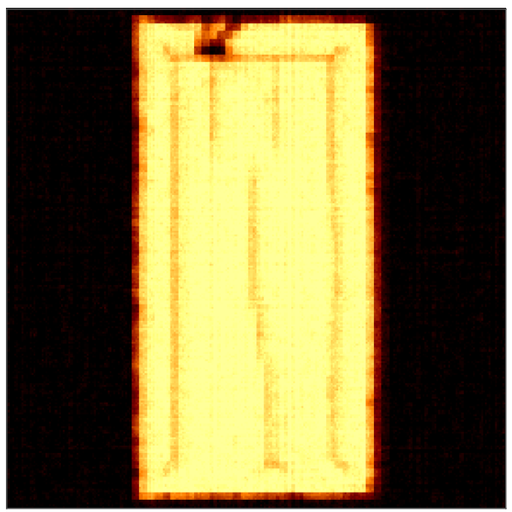
Bad images for UTV | CIGS_0254

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

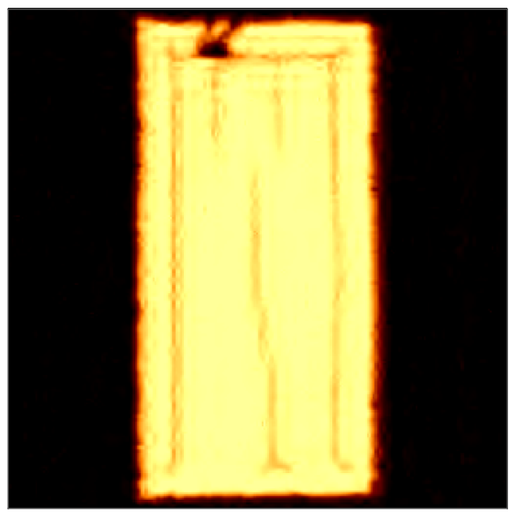
Ground truth
Ground truth



Zero-fill
PSNR 26.23 dB | SSIM 0.507
HaarPSI 0.603



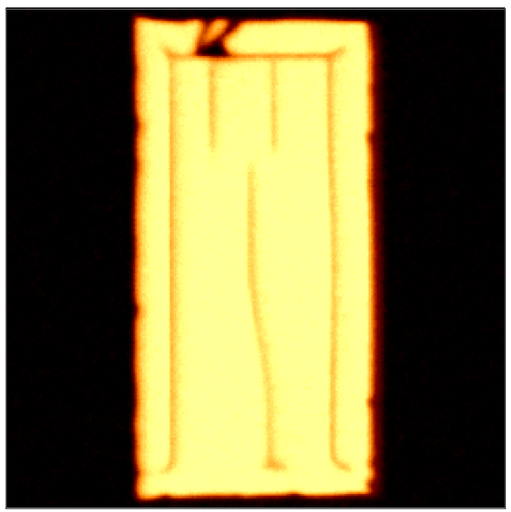
SPGL1-BPDN
PSNR 32.11 dB | SSIM 0.824
HaarPSI 0.764



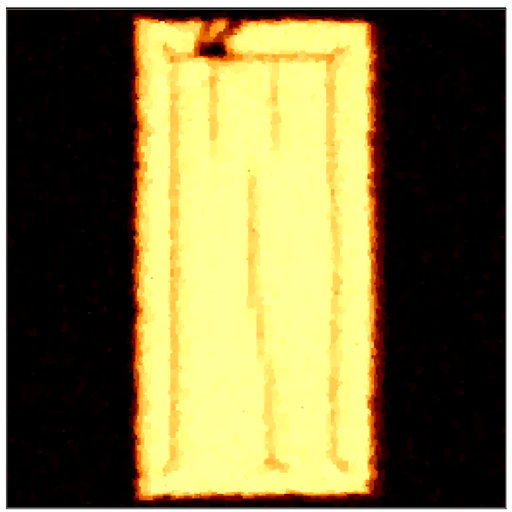
PnP-ADMM-tuned-5
PSNR 39.50 dB | SSIM 0.983
HaarPSI 0.947



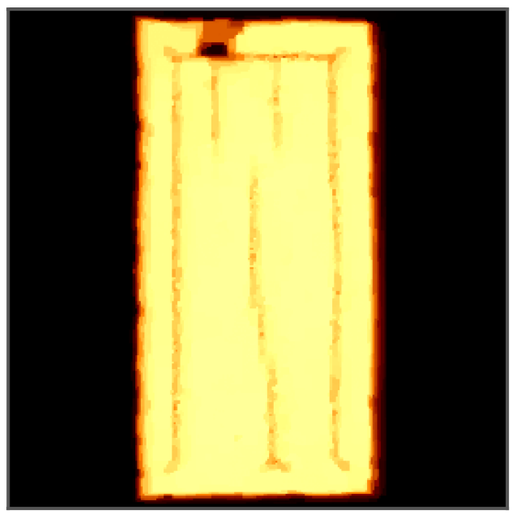
GS-PnP-tuned-5
PSNR 34.12 dB | SSIM 0.806
HaarPSI 0.887



Scalar TV
PSNR 31.72 dB | SSIM 0.766
HaarPSI 0.845



UTV C60
PSNR 33.34 dB | SSIM 0.938
HaarPSI 0.825





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

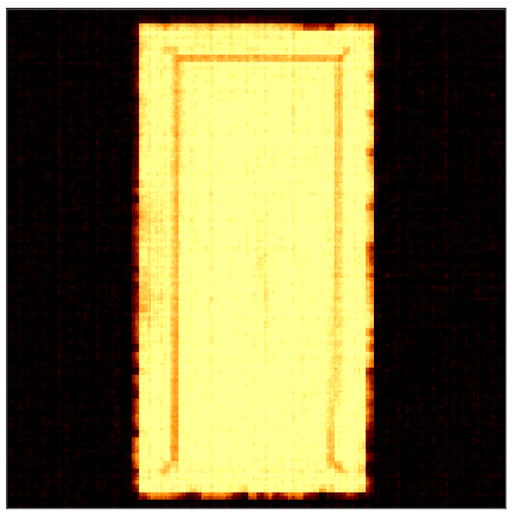
Bad images for UTV | CIGS_0710

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

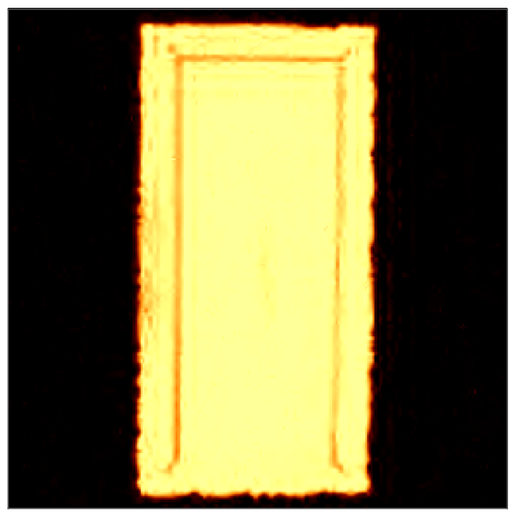
Ground truth
Ground truth



Zero-fill
PSNR 24.80 dB | SSIM 0.411
HaarPSI 0.493



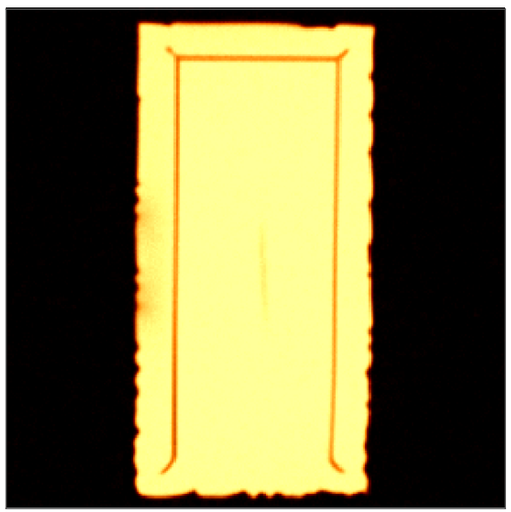
SPGL1-BPDN
PSNR 28.87 dB | SSIM 0.756
HaarPSI 0.587



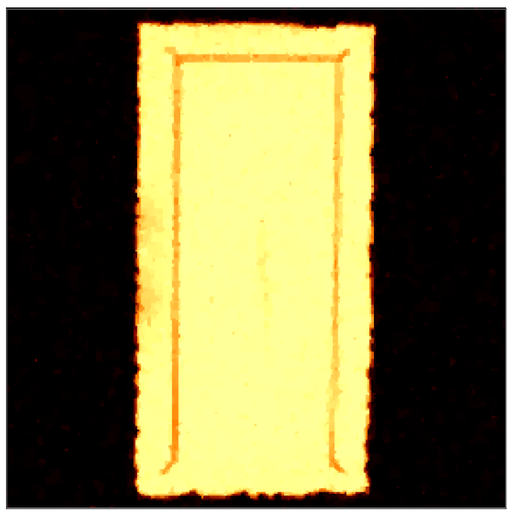
PnP-ADMM-tuned-5
PSNR 37.25 dB | SSIM 0.985
HaarPSI 0.937



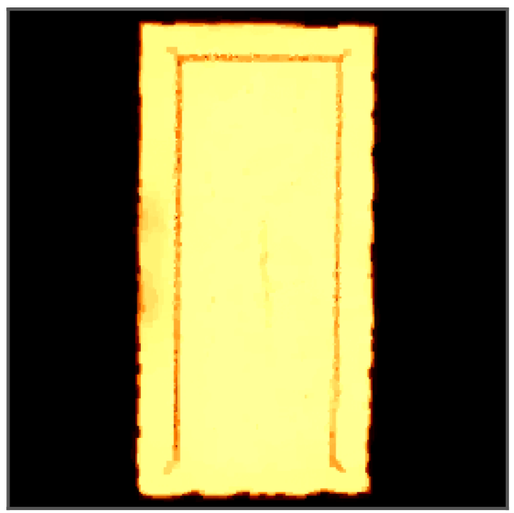
GS-PnP-tuned-5
PSNR 34.66 dB | SSIM 0.864
HaarPSI 0.895



Scalar TV
PSNR 30.48 dB | SSIM 0.738
HaarPSI 0.801



UTV C60
PSNR 32.60 dB | SSIM 0.957
HaarPSI 0.875





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

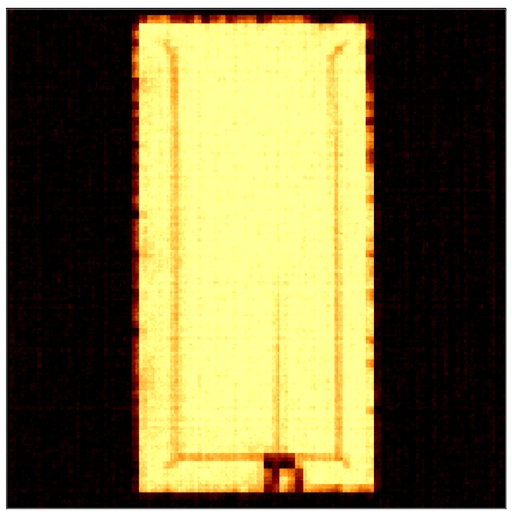
Bad images for UTV | CIGS_0727

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

Ground truth
Ground truth



Zero-fill
PSNR 24.31 dB | SSIM 0.417
HaarPSI 0.489



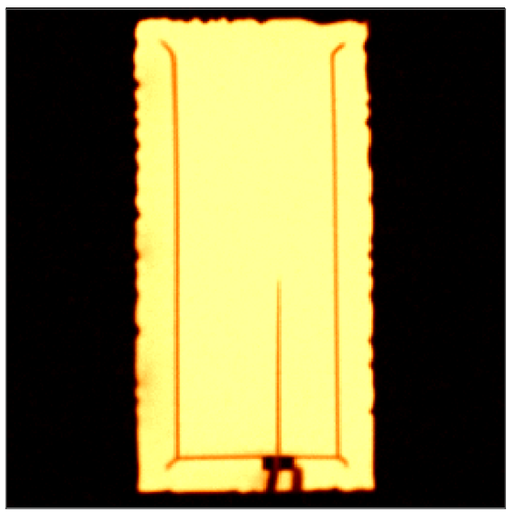
SPGL1-BPDN
PSNR 27.62 dB | SSIM 0.712
HaarPSI 0.555



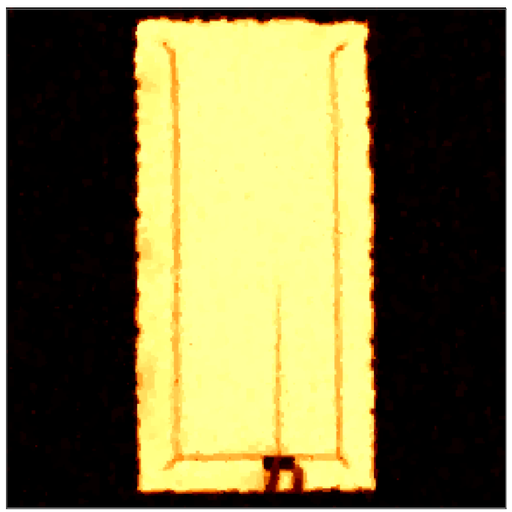
PnP-ADMM-tuned-5
PSNR 36.98 dB | SSIM 0.978
HaarPSI 0.927



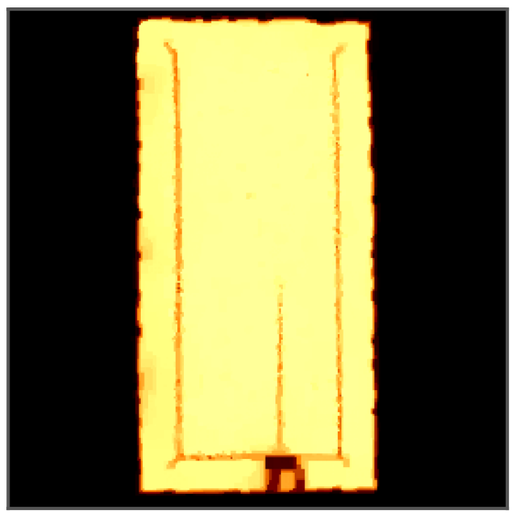
GS-PnP-tuned-5
PSNR 34.46 dB | SSIM 0.865
HaarPSI 0.891



Scalar TV
PSNR 30.12 dB | SSIM 0.726
HaarPSI 0.797



UTV C60
PSNR 31.51 dB | SSIM 0.945
HaarPSI 0.856





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

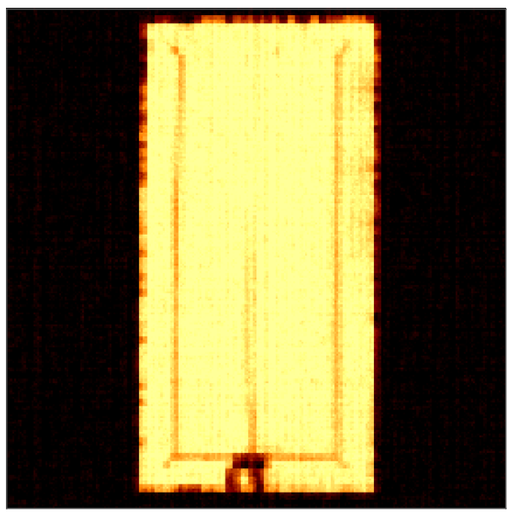
Bad images for UTV | CIGS_0805

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

Ground truth
Ground truth



Zero-fill
PSNR 24.34 dB | SSIM 0.419
HaarPSI 0.500



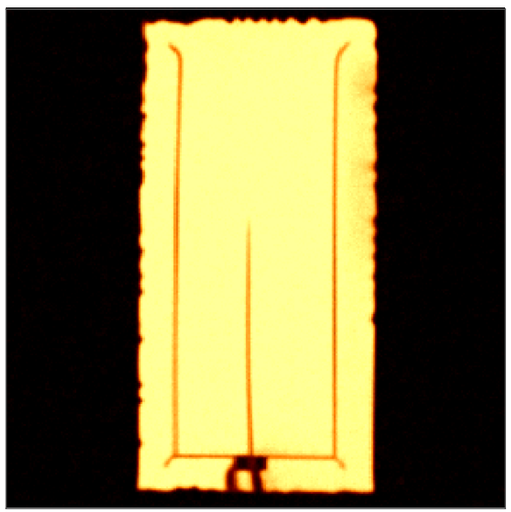
SPGL1-BPDN
PSNR 28.09 dB | SSIM 0.725
HaarPSI 0.554



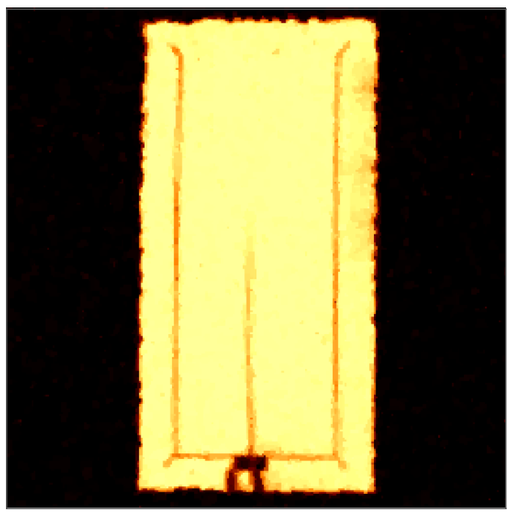
PnP-ADMM-tuned-5
PSNR 36.81 dB | SSIM 0.982
HaarPSI 0.926



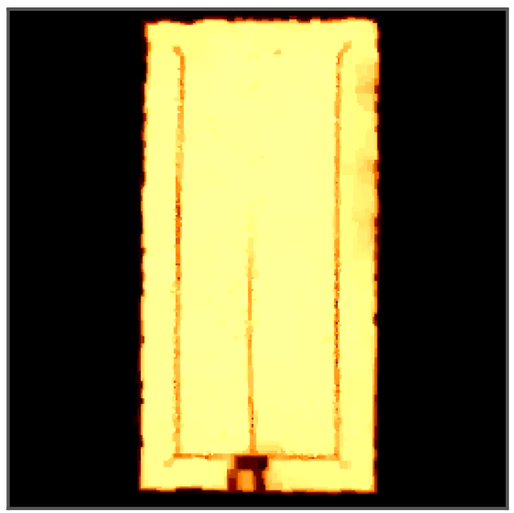
GS-PnP-tuned-5
PSNR 33.19 dB | SSIM 0.790
HaarPSI 0.866



Scalar TV
PSNR 30.41 dB | SSIM 0.745
HaarPSI 0.807



UTV C60
PSNR 31.65 dB | SSIM 0.951
HaarPSI 0.868





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

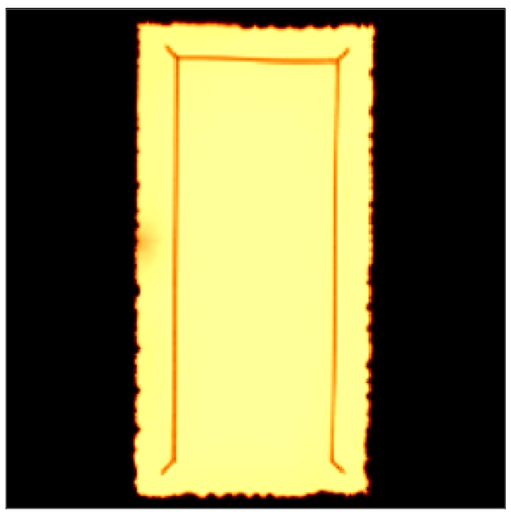
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

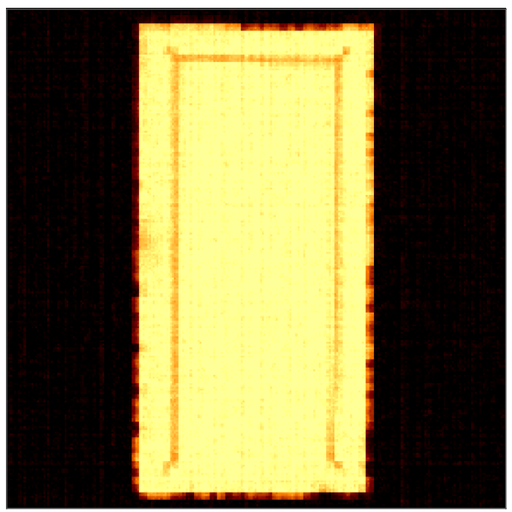
Bad images for UTV | CIGS_0821

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

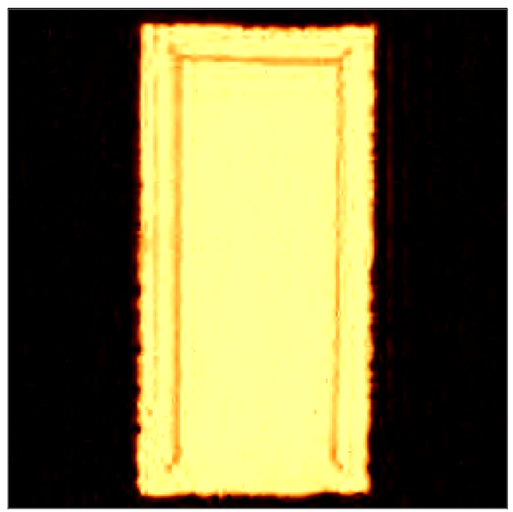
Ground truth
Ground truth



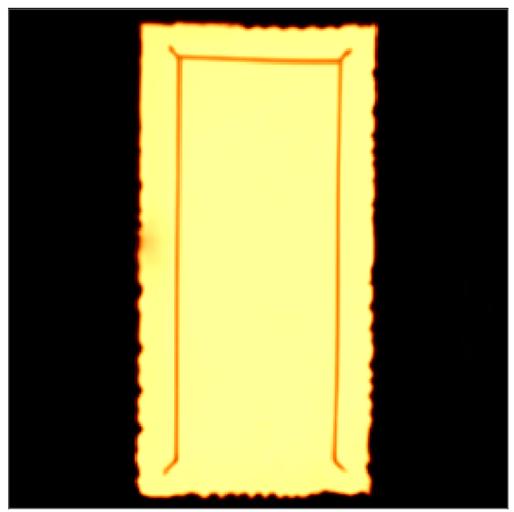
Zero-fill
PSNR 24.52 dB | SSIM 0.403
HaarPSI 0.492



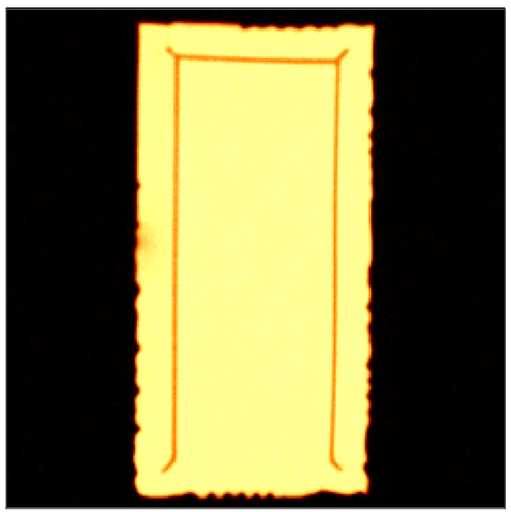
SPGL1-BPDN
PSNR 28.31 dB | SSIM 0.735
HaarPSI 0.536



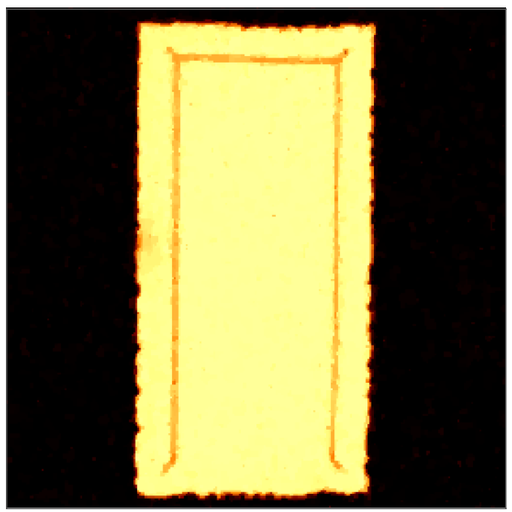
PnP-ADMM-tuned-5
PSNR 37.80 dB | SSIM 0.987
HaarPSI 0.938



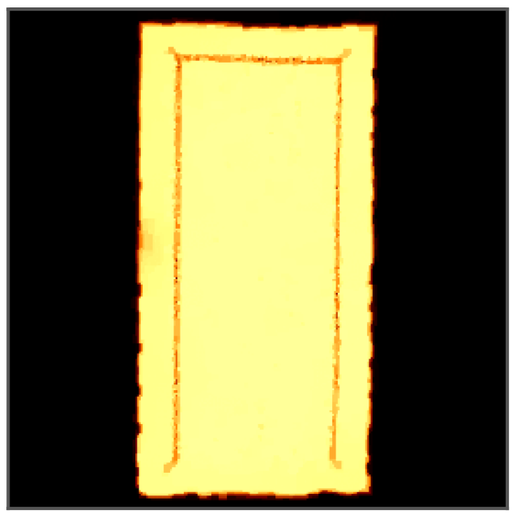
GS-PnP-tuned-5
PSNR 33.60 dB | SSIM 0.791
HaarPSI 0.873



Scalar TV
PSNR 30.95 dB | SSIM 0.741
HaarPSI 0.808



UTV C60
PSNR 32.53 dB | SSIM 0.960
HaarPSI 0.892





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

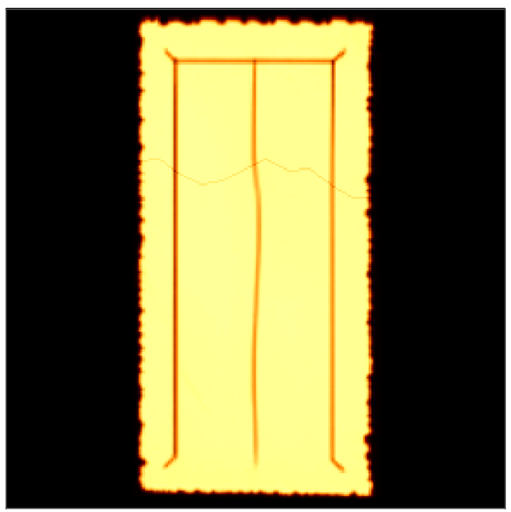
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

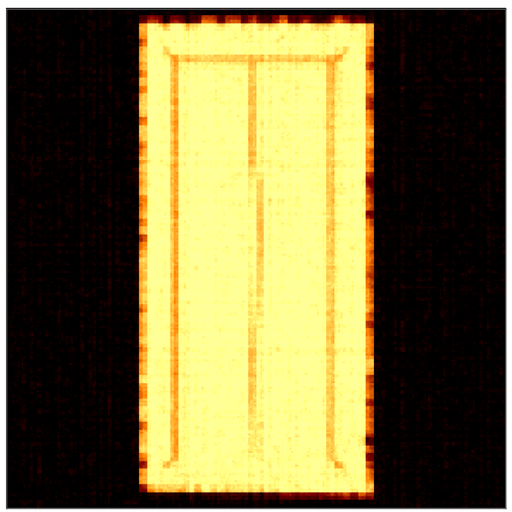
Bad images for UTV | CIGS_0873

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

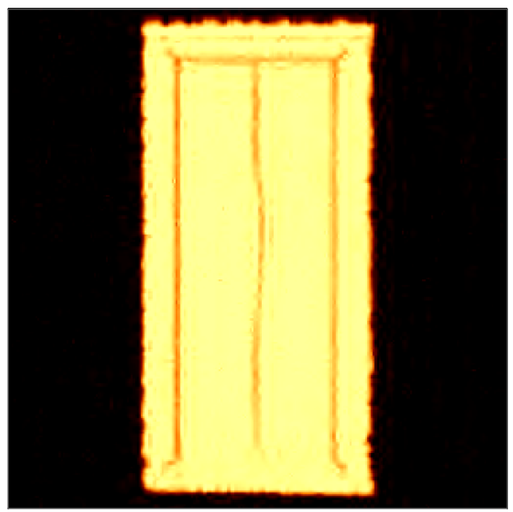
Ground truth
Ground truth



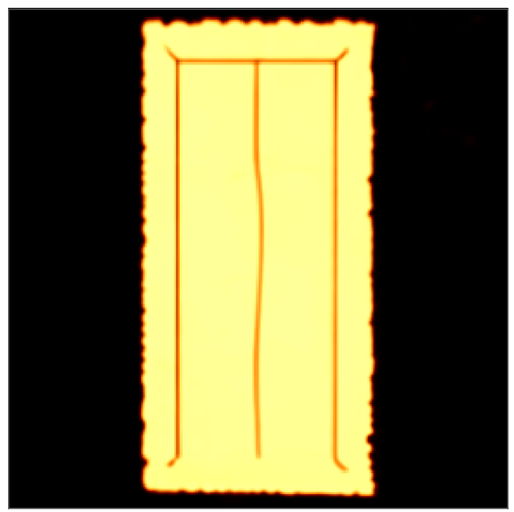
Zero-fill
PSNR 23.61 dB | SSIM 0.416
HaarPSI 0.458



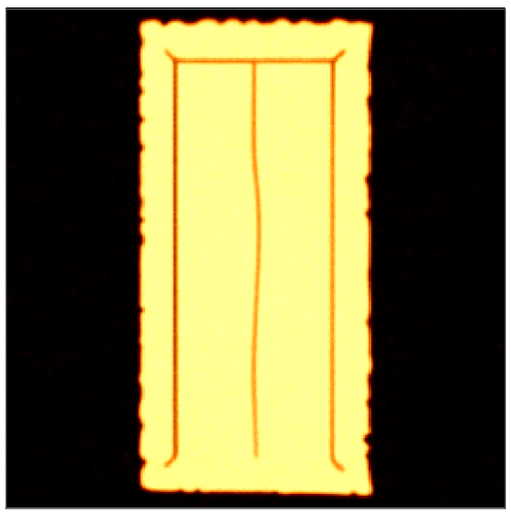
SPGL1-BPDN
PSNR 28.43 dB | SSIM 0.740
HaarPSI 0.591



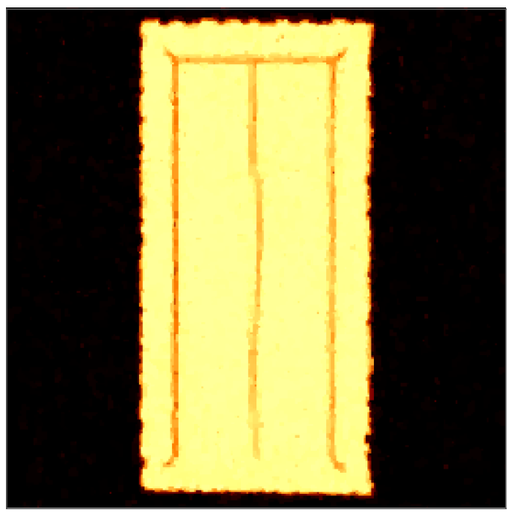
PnP-ADMM-tuned-5
PSNR 37.82 dB | SSIM 0.979
HaarPSI 0.946



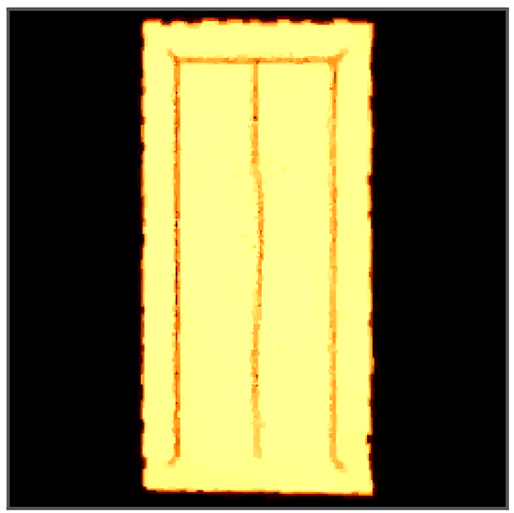
GS-PnP-tuned-5
PSNR 33.39 dB | SSIM 0.789
HaarPSI 0.874



Scalar TV
PSNR 29.47 dB | SSIM 0.727
HaarPSI 0.767



UTV C60
PSNR 31.32 dB | SSIM 0.948
HaarPSI 0.857





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

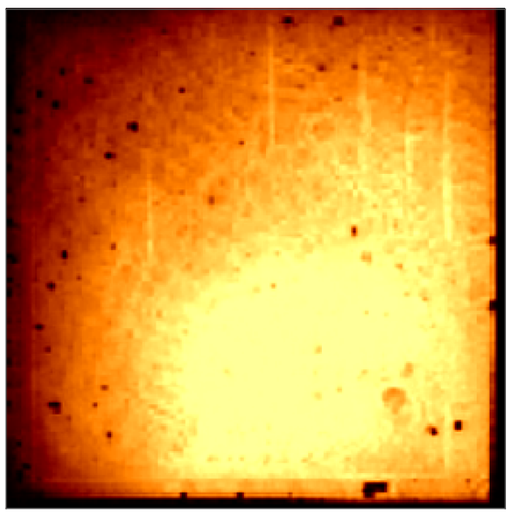
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

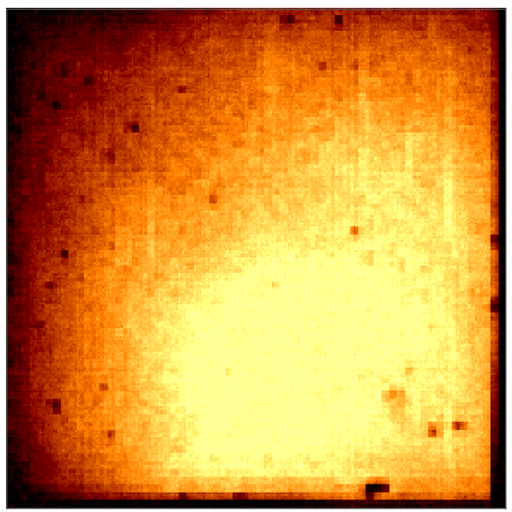
Bad images for UTV | Perovskite_0021

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

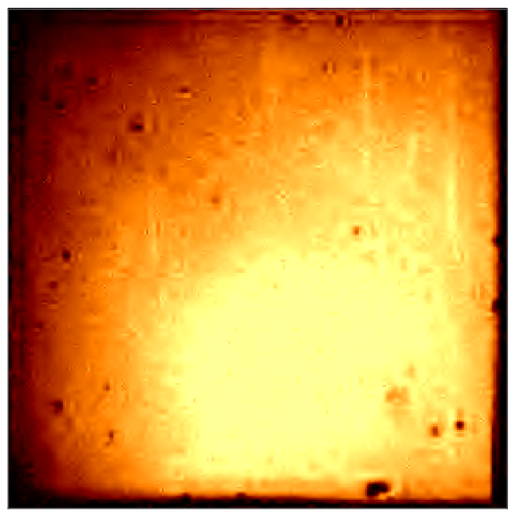
Ground truth
Ground truth



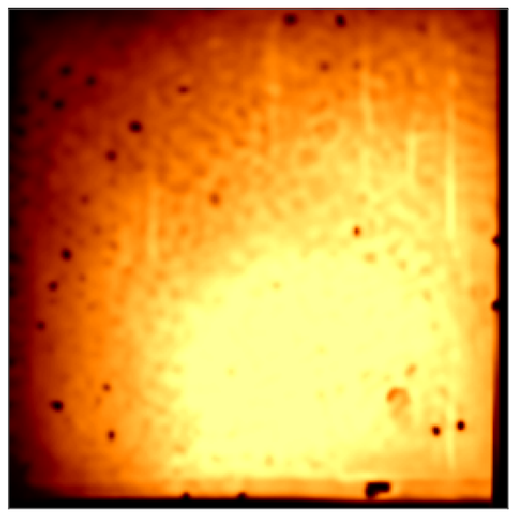
Zero-fill
PSNR 27.54 dB | SSIM 0.619
HaarPSI 0.626



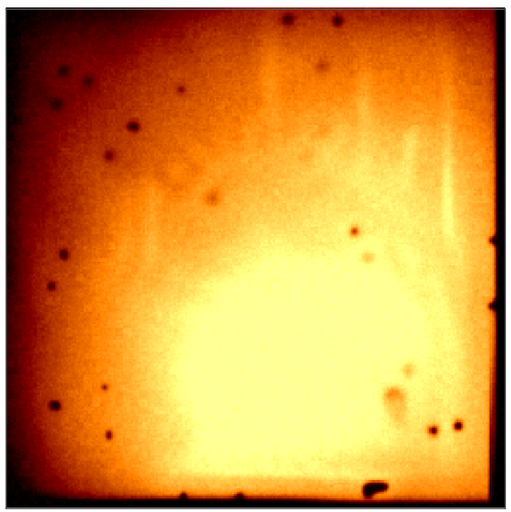
SPGL1-BPDN
PSNR 30.01 dB | SSIM 0.767
HaarPSI 0.662



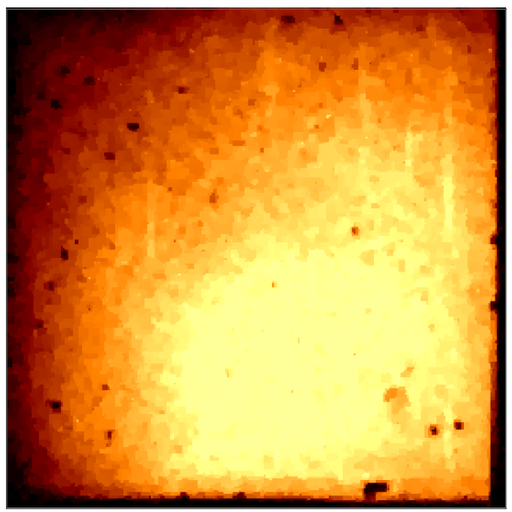
PnP-ADMM-tuned-5
PSNR 33.50 dB | SSIM 0.866
HaarPSI 0.801



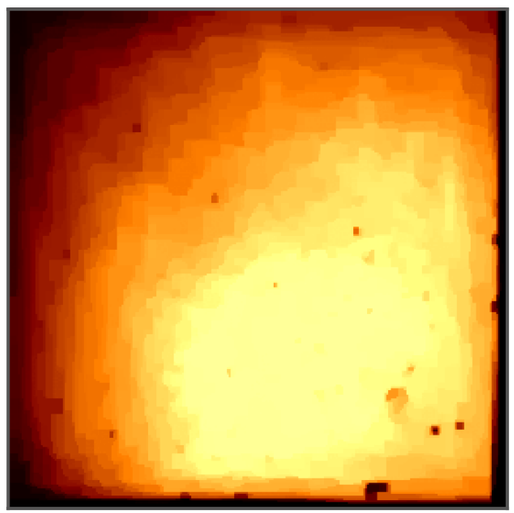
GS-PnP-tuned-5
PSNR 29.97 dB | SSIM 0.668
HaarPSI 0.676

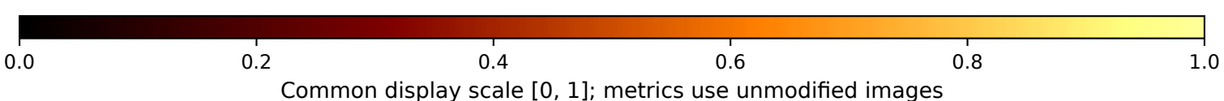


Scalar TV
PSNR 31.04 dB | SSIM 0.785
HaarPSI 0.744



UTV C60
PSNR 30.95 dB | SSIM 0.796
HaarPSI 0.623





Comparison notes

All methods share the exact frozen measurement and sampling indices.

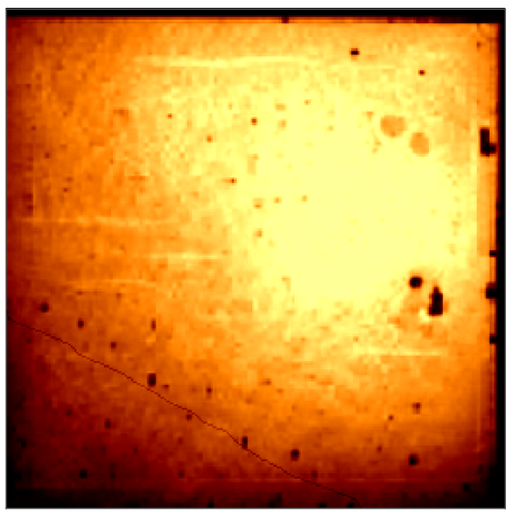
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

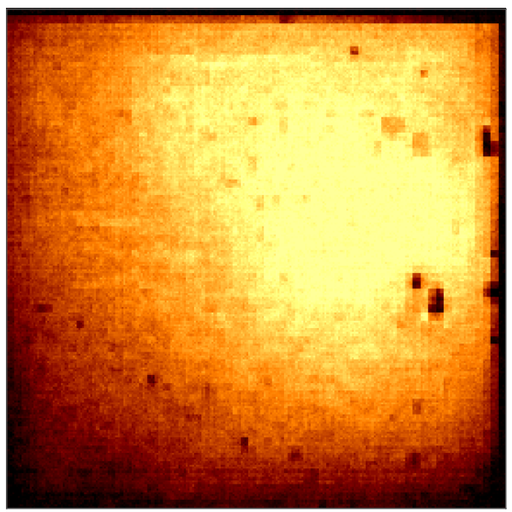
Bad images for UTV | Perovskite_0031

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

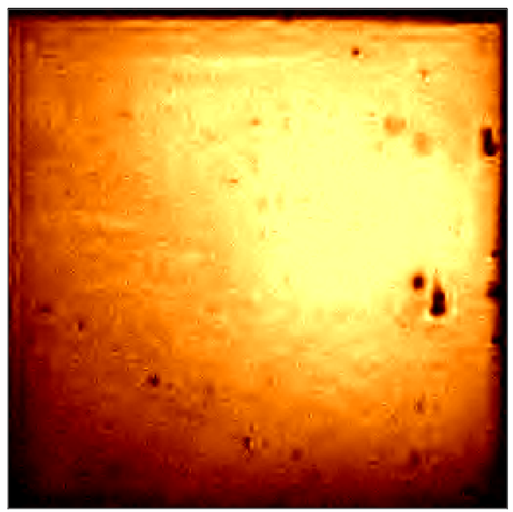
Ground truth
Ground truth



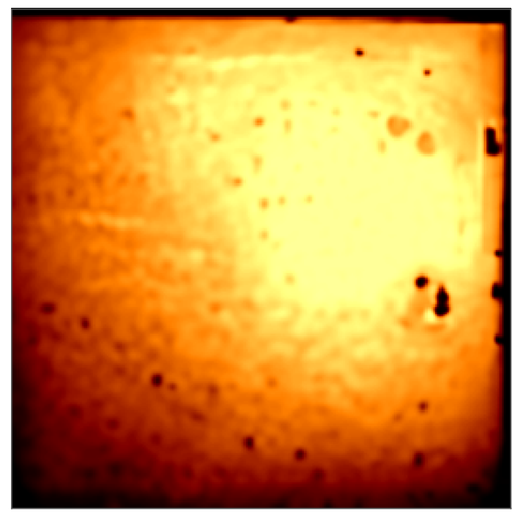
Zero-fill
PSNR 27.19 dB | SSIM 0.606
HaarPSI 0.616



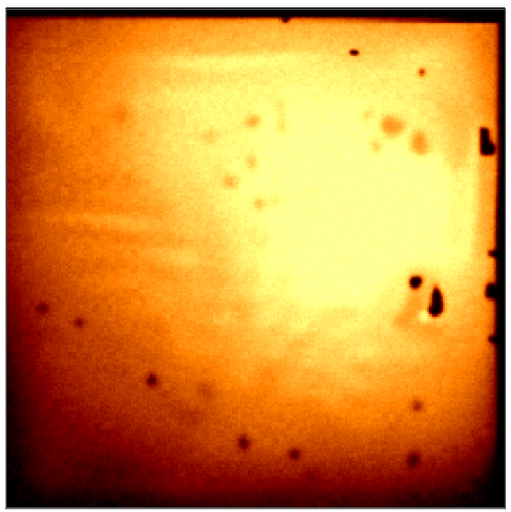
SPGL1-BPDN
PSNR 29.82 dB | SSIM 0.751
HaarPSI 0.685



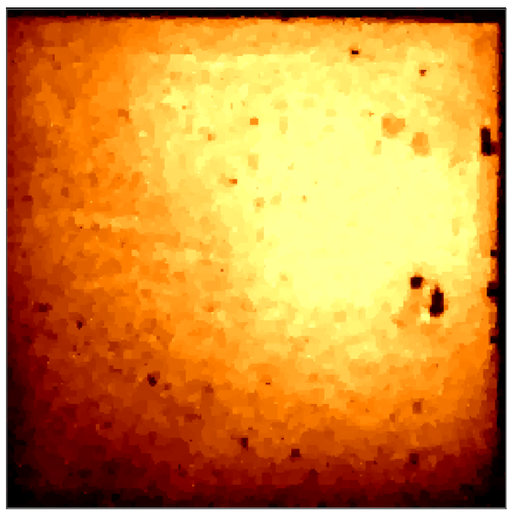
PnP-ADMM-tuned-5
PSNR 33.00 dB | SSIM 0.849
HaarPSI 0.789



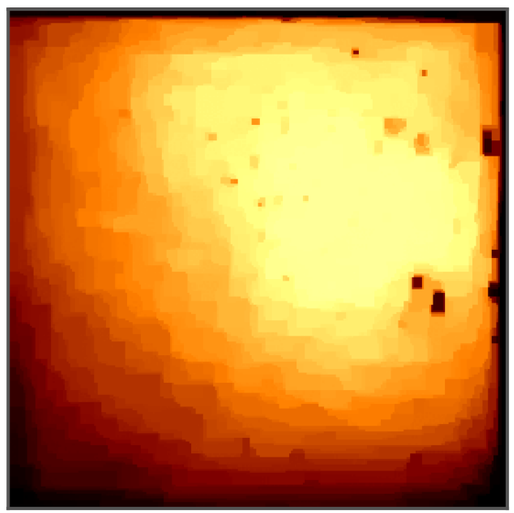
GS-PnP-tuned-5
PSNR 29.68 dB | SSIM 0.652
HaarPSI 0.662



Scalar TV
PSNR 30.68 dB | SSIM 0.773
HaarPSI 0.745



UTV C60
PSNR 30.46 dB | SSIM 0.780
HaarPSI 0.615





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

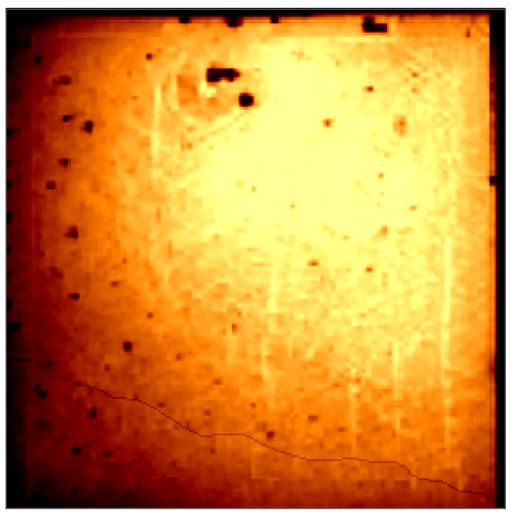
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

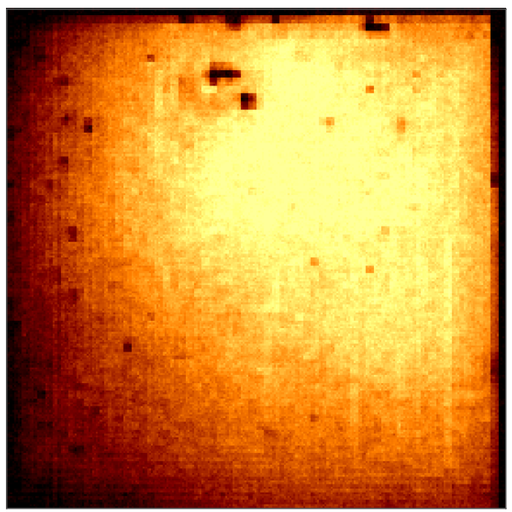
Bad images for UTV | Perovskite_0079

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

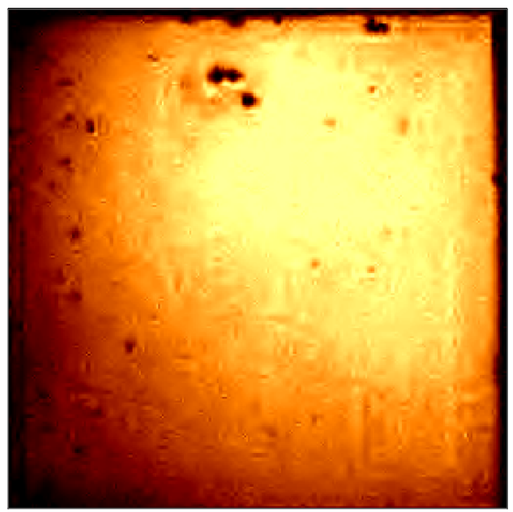
Ground truth
Ground truth



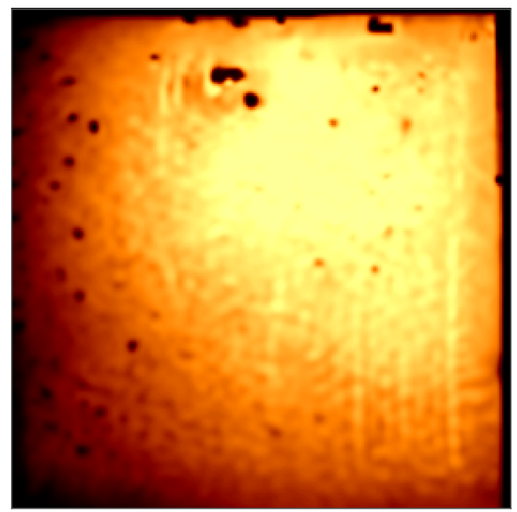
Zero-fill
PSNR 27.03 dB | SSIM 0.608
HaarPSI 0.618



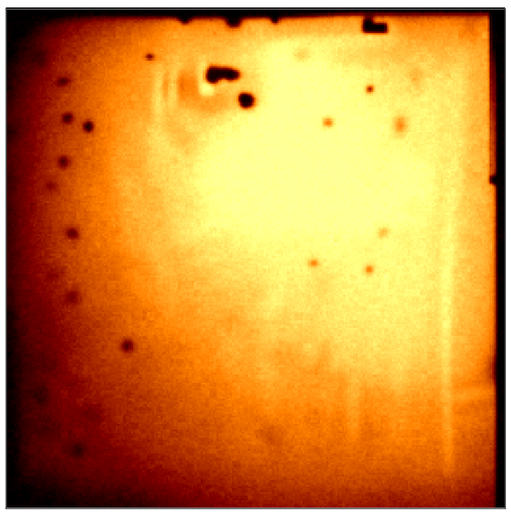
SPGL1-BPDN
PSNR 29.42 dB | SSIM 0.746
HaarPSI 0.661



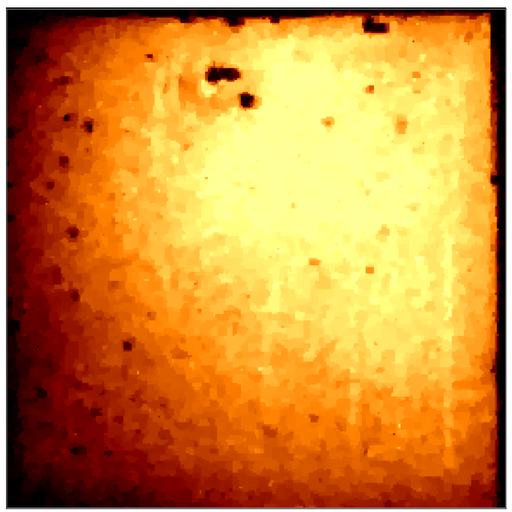
PnP-ADMM-tuned-5
PSNR 32.73 dB | SSIM 0.846
HaarPSI 0.787



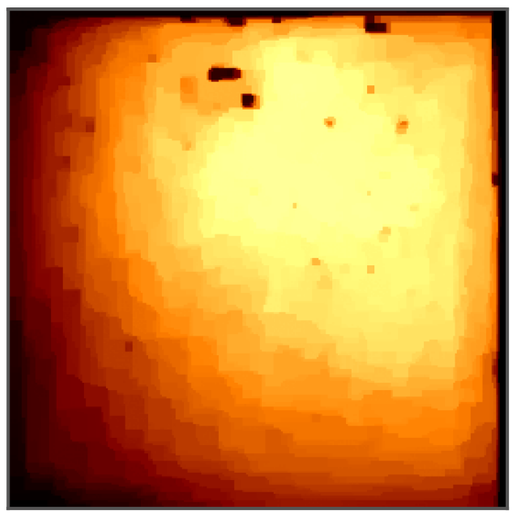
GS-PnP-tuned-5
PSNR 29.45 dB | SSIM 0.640
HaarPSI 0.661



Scalar TV
PSNR 30.44 dB | SSIM 0.771
HaarPSI 0.729



UTV C60
PSNR 29.98 dB | SSIM 0.762
HaarPSI 0.613





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

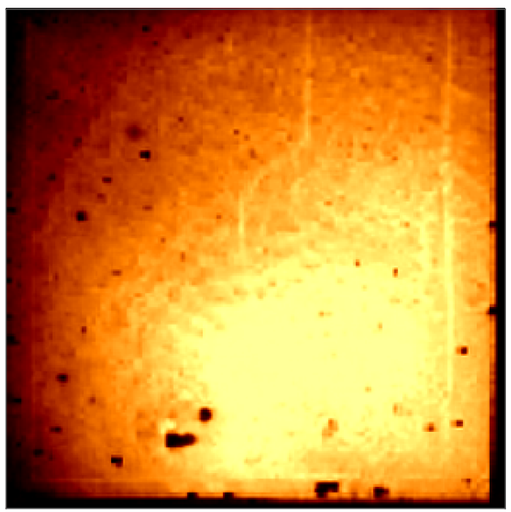
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

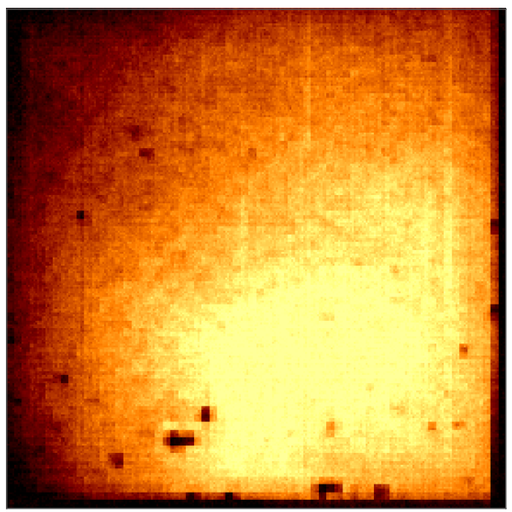
Bad images for UTV | Perovskite_0098

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

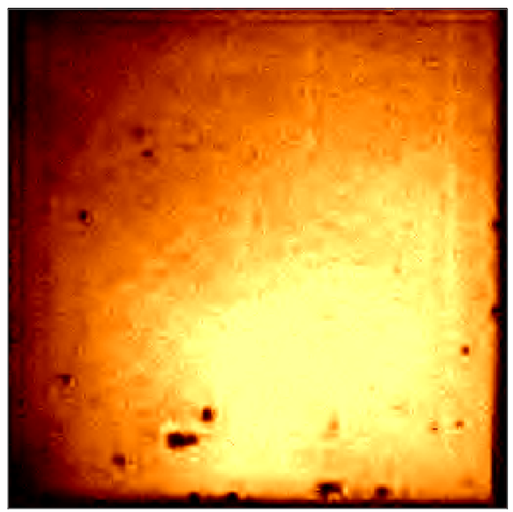
Ground truth
Ground truth



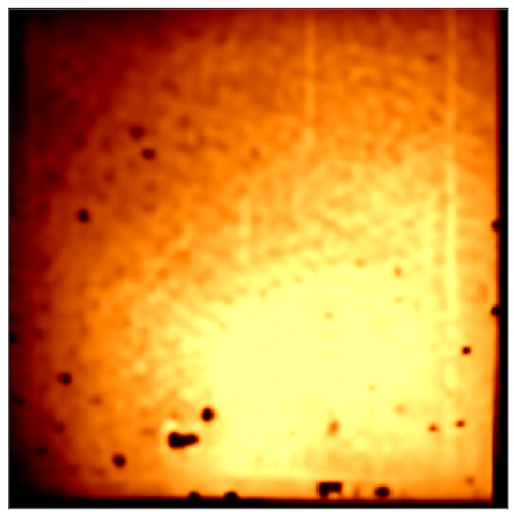
Zero-fill
PSNR 27.22 dB | SSIM 0.608
HaarPSI 0.624



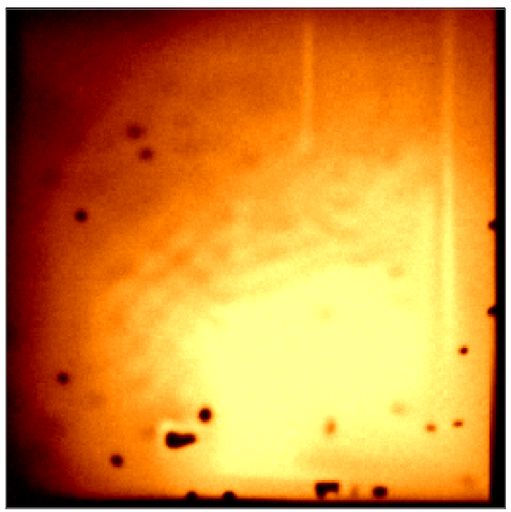
SPGL1-BPDN
PSNR 29.65 dB | SSIM 0.755
HaarPSI 0.646



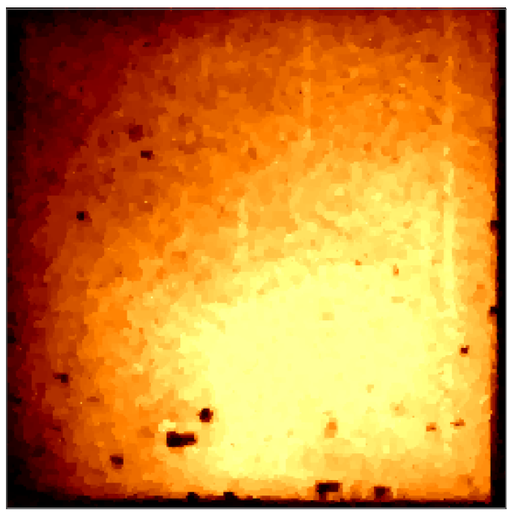
PnP-ADMM-tuned-5
PSNR 32.91 dB | SSIM 0.854
HaarPSI 0.785



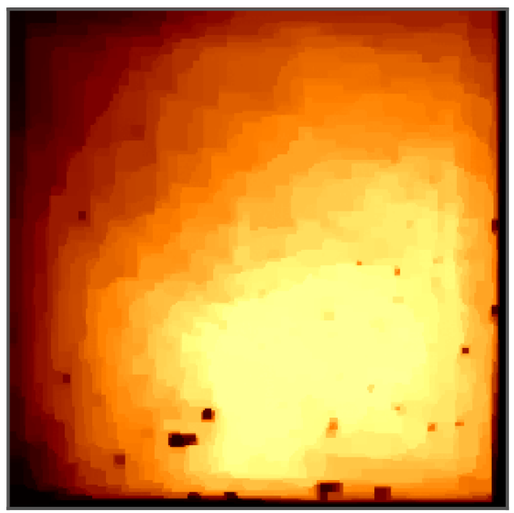
GS-PnP-tuned-5
PSNR 30.52 dB | SSIM 0.715
HaarPSI 0.684



Scalar TV
PSNR 30.72 dB | SSIM 0.781
HaarPSI 0.729



UTV C60
PSNR 30.46 dB | SSIM 0.780
HaarPSI 0.616





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

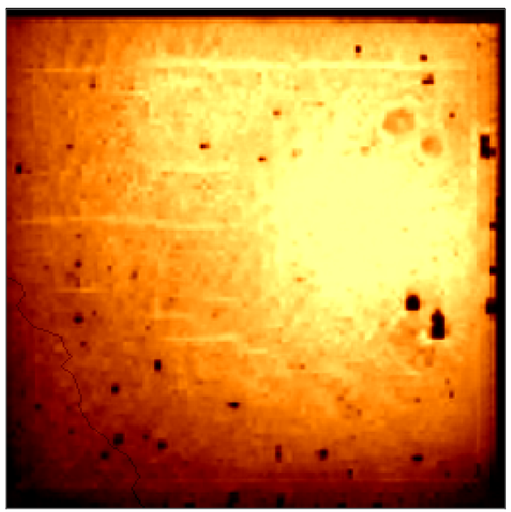
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

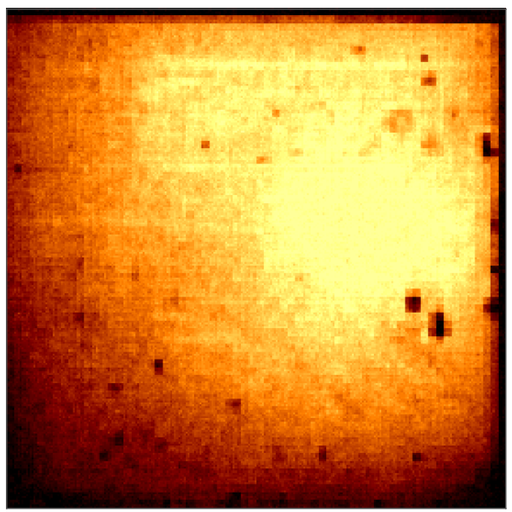
Bad images for UTV | Perovskite_0187

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

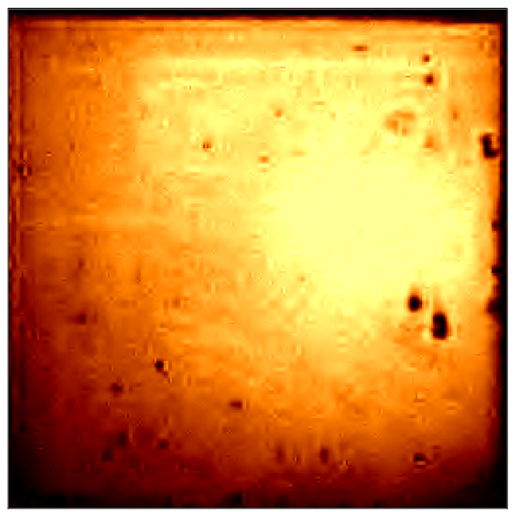
Ground truth
Ground truth



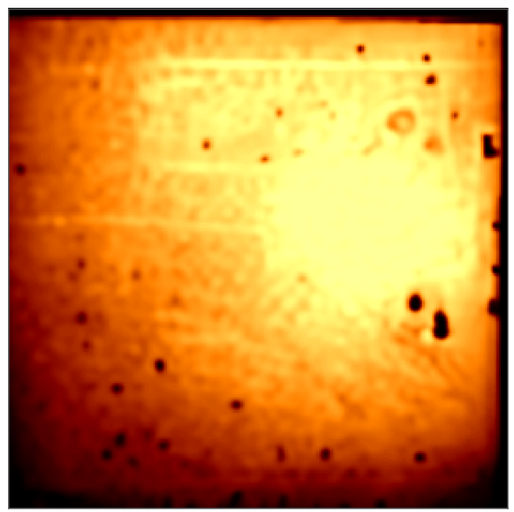
Zero-fill
PSNR 26.88 dB | SSIM 0.615
HaarPSI 0.612



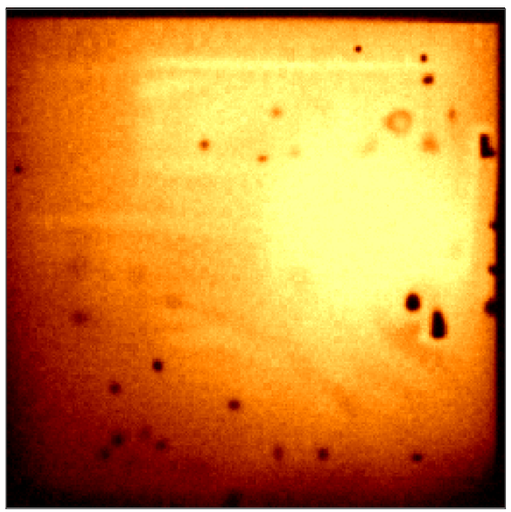
SPGL1-BPDN
PSNR 29.25 dB | SSIM 0.745
HaarPSI 0.666



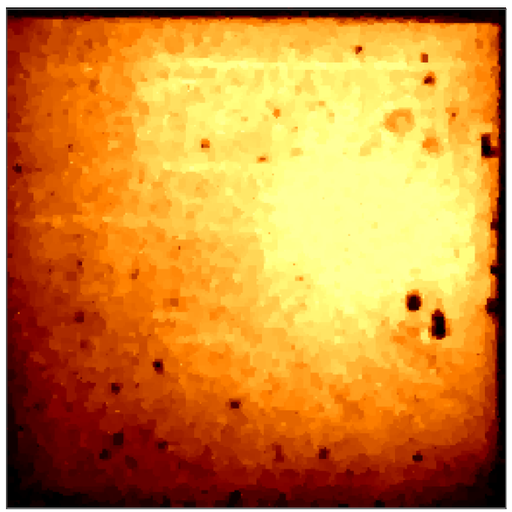
PnP-ADMM-tuned-5
PSNR 32.65 dB | SSIM 0.850
HaarPSI 0.786



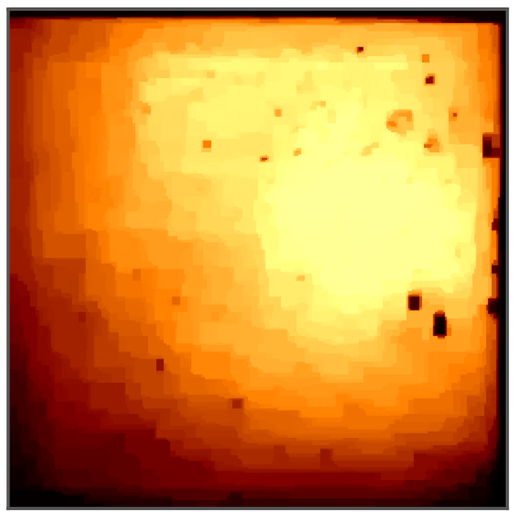
GS-PnP-tuned-5
PSNR 28.50 dB | SSIM 0.584
HaarPSI 0.636



Scalar TV
PSNR 30.43 dB | SSIM 0.776
HaarPSI 0.743



UTV C60
PSNR 29.82 dB | SSIM 0.766
HaarPSI 0.603





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

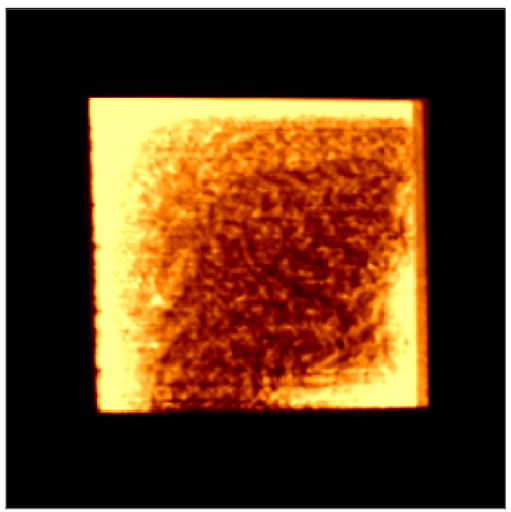
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

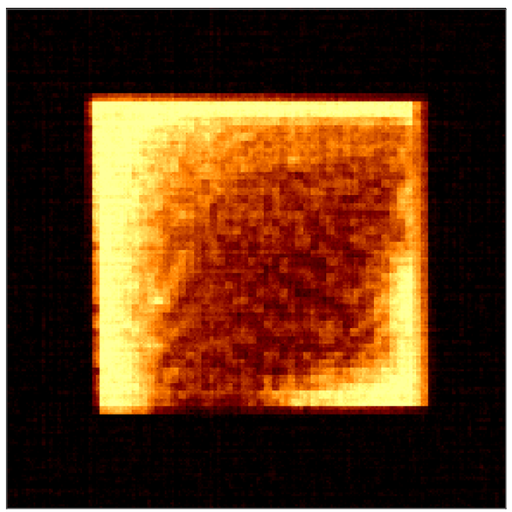
Bad images for UTV | Perovskite_0260

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

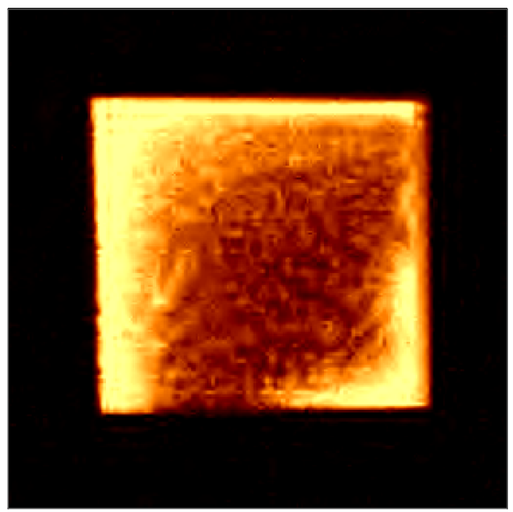
Ground truth
Ground truth



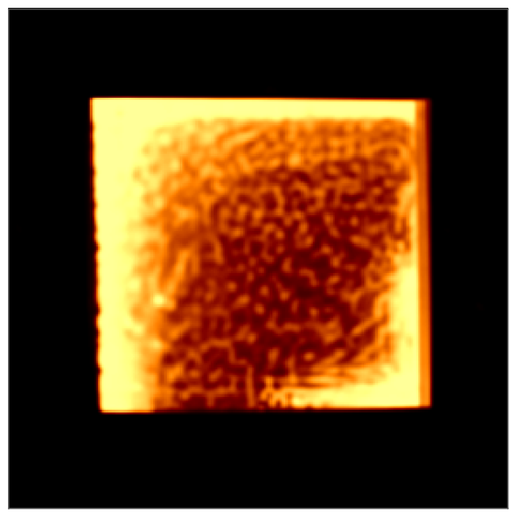
Zero-fill
PSNR 24.51 dB | SSIM 0.473
HaarPSI 0.532



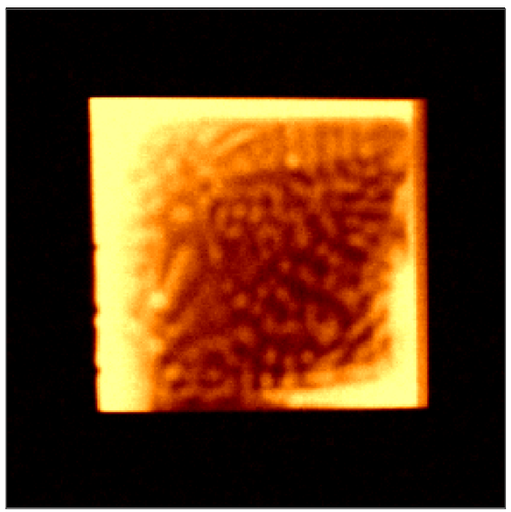
SPGL1-BPDN
PSNR 27.66 dB | SSIM 0.709
HaarPSI 0.580



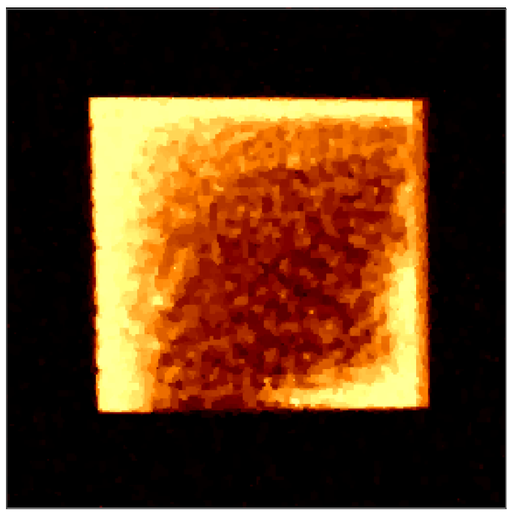
PnP-ADMM-tuned-5
PSNR 32.27 dB | SSIM 0.928
HaarPSI 0.815



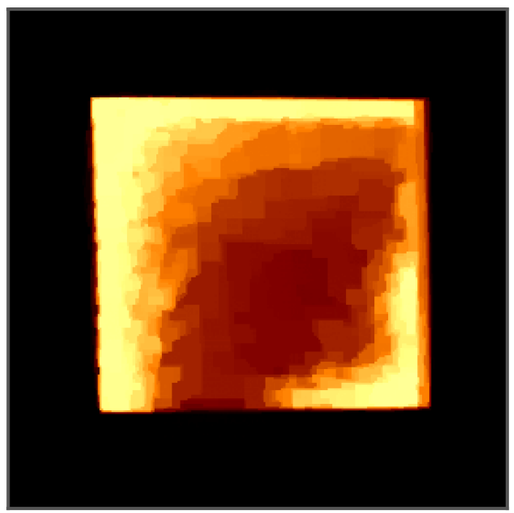
GS-PnP-tuned-5
PSNR 29.34 dB | SSIM 0.711
HaarPSI 0.728



Scalar TV
PSNR 29.79 dB | SSIM 0.731
HaarPSI 0.748



UTV C60
PSNR 28.53 dB | SSIM 0.806
HaarPSI 0.554





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

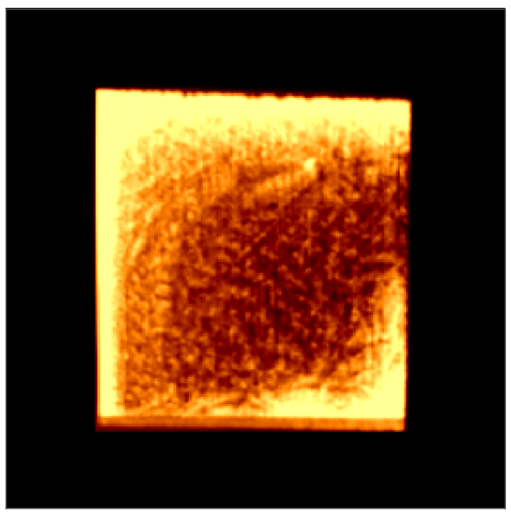
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

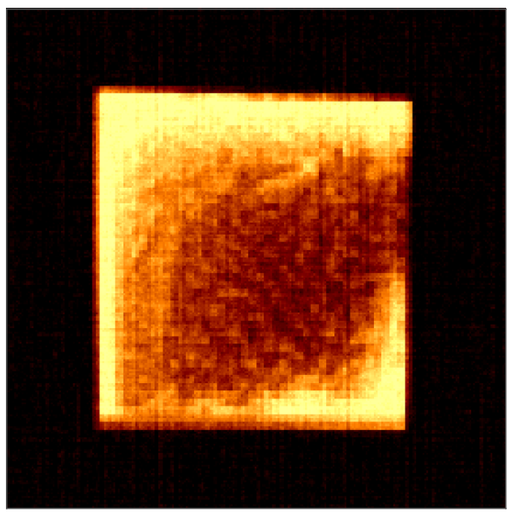
Bad images for UTV | Perovskite_0288

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

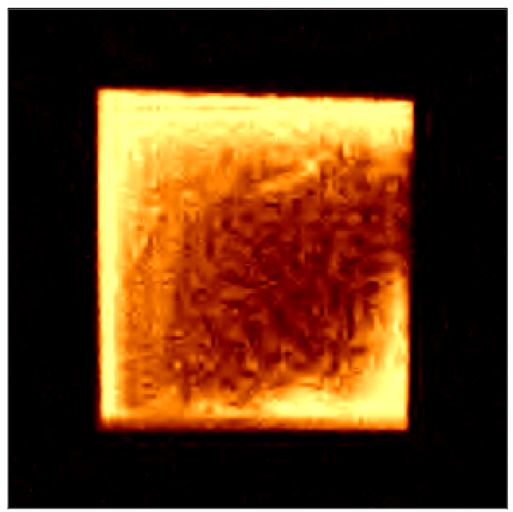
Ground truth
Ground truth



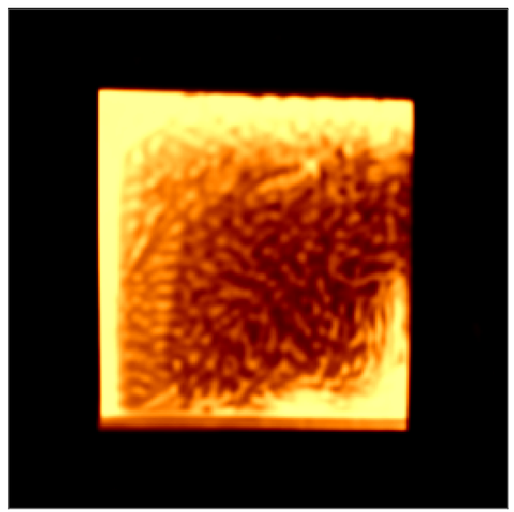
Zero-fill
PSNR 24.69 dB | SSIM 0.464
HaarPSI 0.544



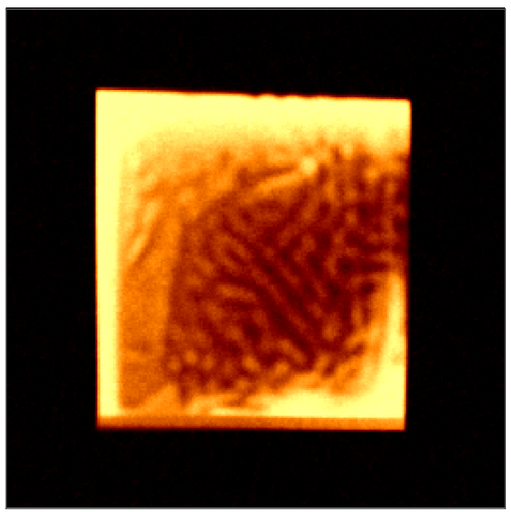
SPGL1-BPDN
PSNR 28.27 dB | SSIM 0.706
HaarPSI 0.605



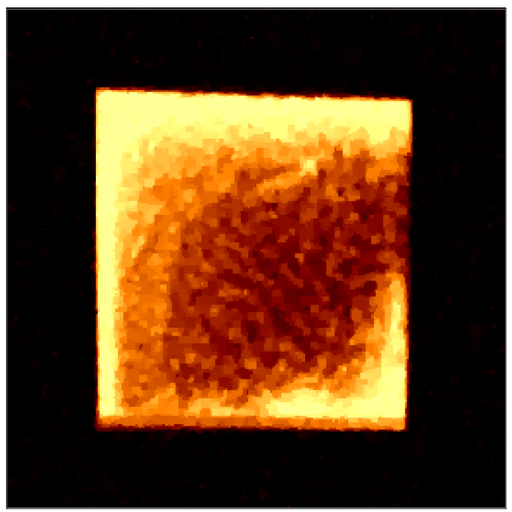
PnP-ADMM-tuned-5
PSNR 32.44 dB | SSIM 0.927
HaarPSI 0.818



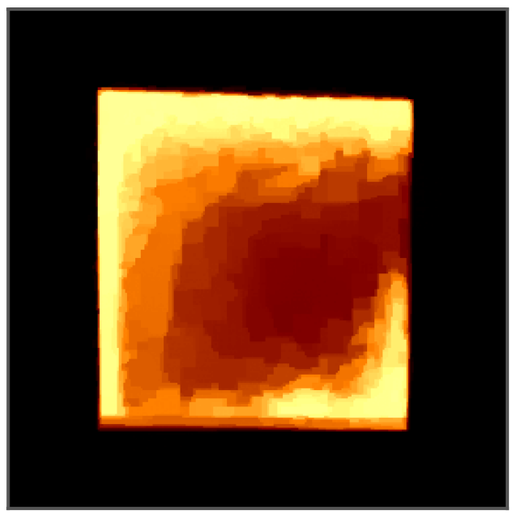
GS-PnP-tuned-5
PSNR 29.27 dB | SSIM 0.698
HaarPSI 0.725



Scalar TV
PSNR 29.88 dB | SSIM 0.726
HaarPSI 0.744



UTV C60
PSNR 28.70 dB | SSIM 0.803
HaarPSI 0.581





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

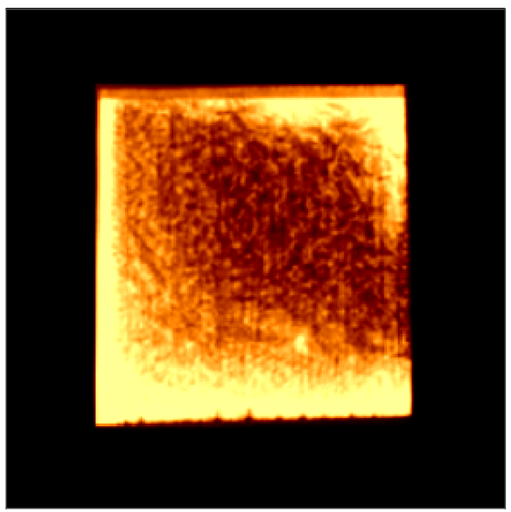
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

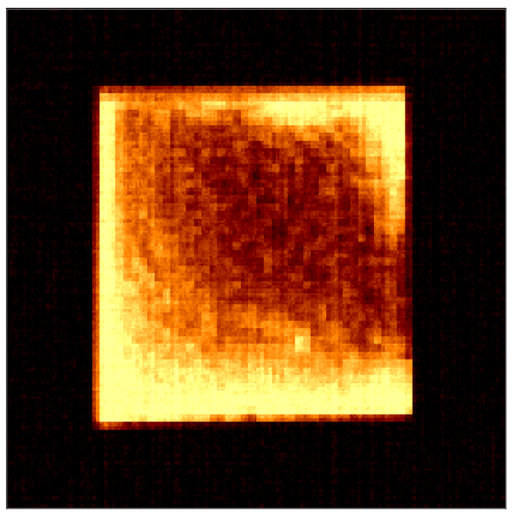
Bad images for UTV | Perovskite_0292

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

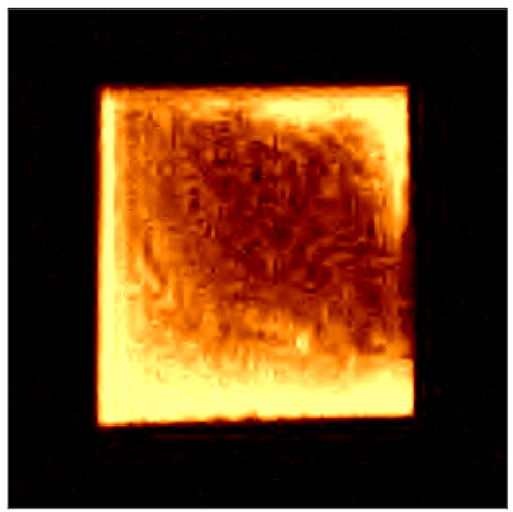
Ground truth
Ground truth



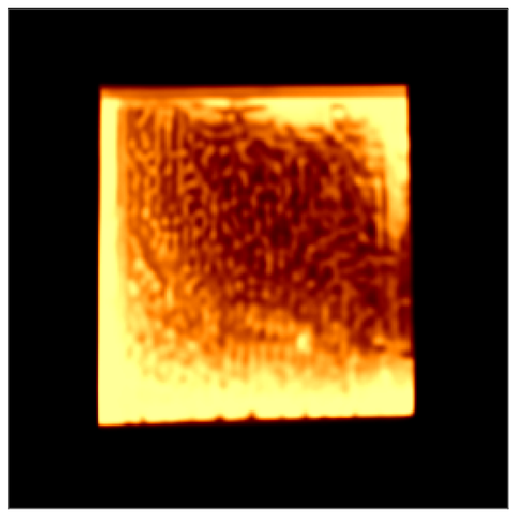
Zero-fill
PSNR 24.57 dB | SSIM 0.476
HaarPSI 0.528



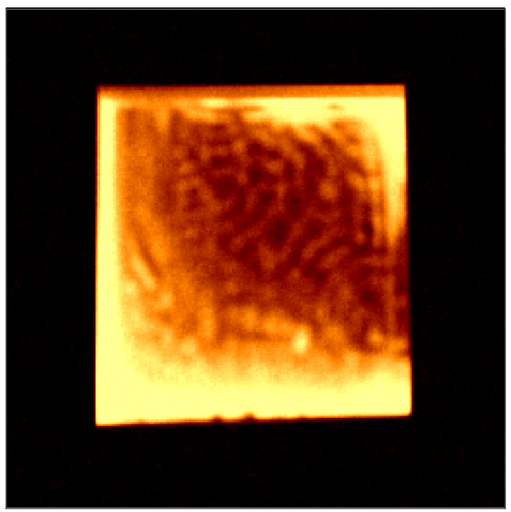
SPGL1-BPDN
PSNR 27.71 dB | SSIM 0.702
HaarPSI 0.591



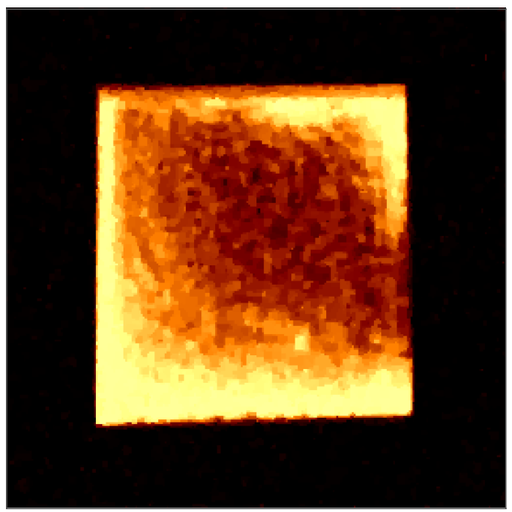
PnP-ADMM-tuned-5
PSNR 32.37 dB | SSIM 0.927
HaarPSI 0.818



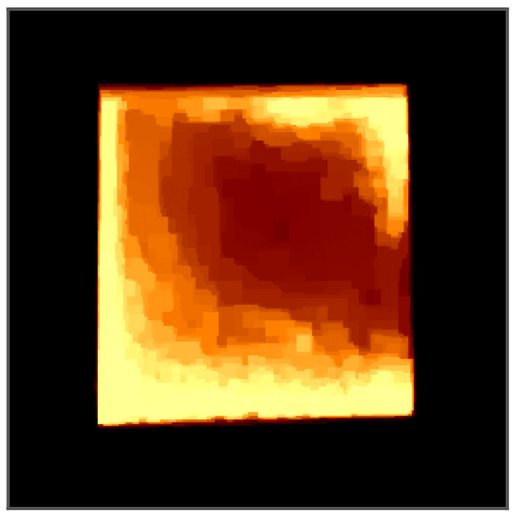
GS-PnP-tuned-5
PSNR 29.20 dB | SSIM 0.688
HaarPSI 0.725



Scalar TV
PSNR 29.79 dB | SSIM 0.717
HaarPSI 0.757



UTV C60
PSNR 28.81 dB | SSIM 0.809
HaarPSI 0.580





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

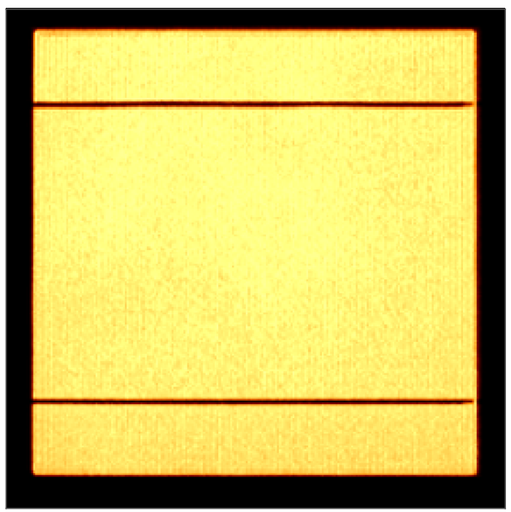
Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Review loss of thin structures, contrast and local details.

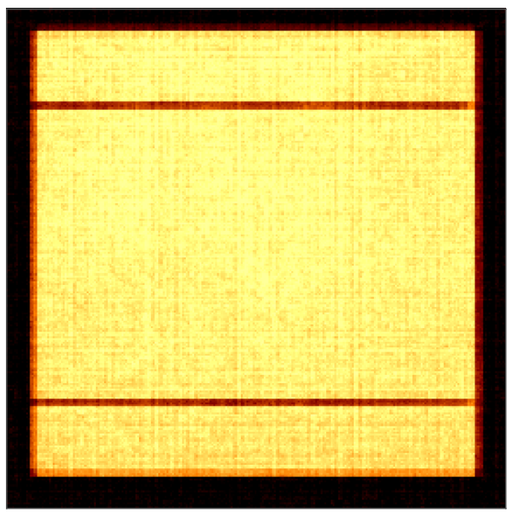
Bad images for UTV | Silicone_0767

Stored 20% multilevel Hadamard sampling | NPL-default noise | identical acquisition

Ground truth
Ground truth



Zero-fill
PSNR 22.40 dB | SSIM 0.450
HaarPSI 0.490



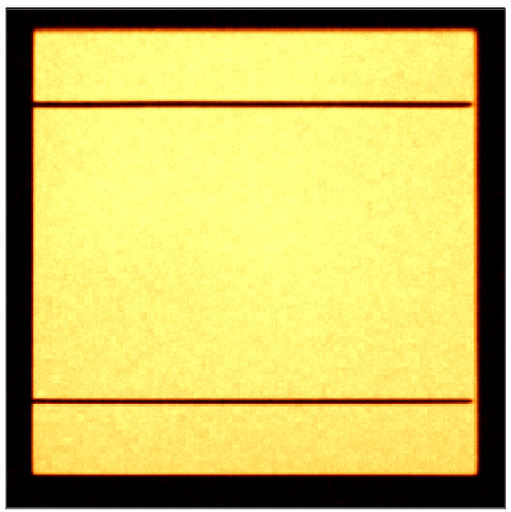
SPGL1-BPDN
PSNR 24.36 dB | SSIM 0.577
HaarPSI 0.503



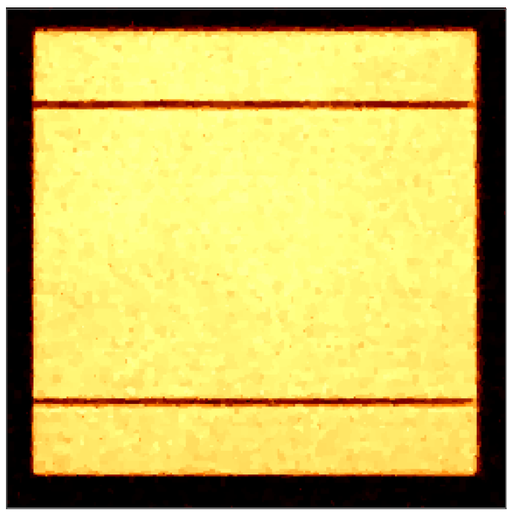
PnP-ADMM-tuned-5
PSNR 33.59 dB | SSIM 0.795
HaarPSI 0.862



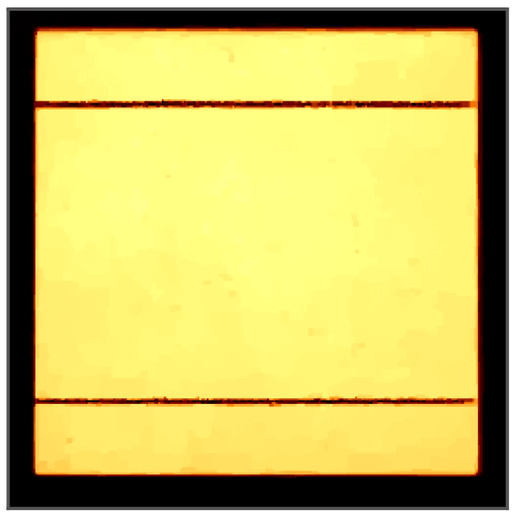
GS-PnP-tuned-5
PSNR 30.66 dB | SSIM 0.622
HaarPSI 0.823



Scalar TV
PSNR 27.78 dB | SSIM 0.675
HaarPSI 0.781



UTV C60
PSNR 31.53 dB | SSIM 0.781
HaarPSI 0.814





Common display scale [0, 1]; metrics use unmodified images

Comparison notes

All methods share the exact frozen measurement and sampling indices.

Scalar TV: lambda=0.011, 1024 iterations.
UTV: original C60 checkpoint.
PnP variants: frozen SSDU-5 choices.

Visual observation: vertical lines remain in zero-fill but are lost by the other methods.